Relations between the functions of one angle

Know one of the four and you know them all — up to a sign the quadrant supplies.

LESSON 49–50 | 2 × 40 MIN | ALGEBRA 9, §21 | IGX 15.5

LESSON CLOCK

13' | 20' | 23' | 20' | 4'

The three relations	13 min
Deriving the second and third	20 min
From one to all four	23 min
Expressing one by another	20 min
Homework	4 min

LEARNING OBJECTIVES

- State the three Pythagorean relations and derive the second and third from the first.
- Use tan α · cot α = 1 and the quotient definitions fluently.
- Find all four functions from any one of them plus a quadrant.
- Express one function in terms of another with the correct sign.

TEXTBOOK

National	pp. 114–119
Cambridge	Core & Extended, pp. 328–333
Workbook	Exercise 15.5

01 Explanation

The three relations

$$\sin^2\alpha + \cos^2\alpha = 1 \quad 1 + \tan^2\alpha = \frac{1}{\cos^2\alpha} \quad 1 + \cot^2\alpha = \frac{1}{\sin^2\alpha}$$

Together with $\tan\alpha = \dfrac{\sin\alpha}{\cos\alpha}$, $\cot\alpha = \dfrac{\cos\alpha}{\sin\alpha}$ and $\tan\alpha \cdot \cot\alpha = 1$, these are the whole of this lesson.

RELATION	VALID FOR
$\sin^2\alpha + \cos^2\alpha = 1$	every α
$1 + \tan^2\alpha = \dfrac{1}{\cos^2\alpha}$	$\cos\alpha \neq 0$
$1 + \cot^2\alpha = \dfrac{1}{\sin^2\alpha}$	$\sin\alpha \neq 0$
$\tan\alpha \cdot \cot\alpha = 1$	both defined

ONLY THE FIRST IS NEW

The second and third are the first, divided. That is worth knowing, because a formula you can rebuild in ten seconds never has to be trusted to memory — and in an examination it never has to be looked up.

Deriving the second and third

Start from $\sin^2\alpha + \cos^2\alpha = 1$ and divide every term by $\cos^2\alpha$:

$$\frac{\sin^2\alpha}{\cos^2\alpha} + \frac{\cos^2\alpha}{\cos^2\alpha} = \frac{1}{\cos^2\alpha} \Longrightarrow \tan^2\alpha + 1 = \frac{1}{\cos^2\alpha}$$

Divide instead by $\sin^2\alpha$ and the third relation appears the same way:

$$1 + \cot^2\alpha = \frac{1}{\sin^2\alpha}$$

DIVIDING BY SOMETHING THAT MAY BE ZERO COSTS THE IDENTITY A CONDITION

The first relation holds everywhere. The second is meaningless when $\cos\alpha = 0$ — at $\alpha = \dfrac{\pi}{2}$, $\tan\alpha$ does not exist and neither does $\dfrac{1}{\cos^2\alpha}$. Writing the condition beside the formula is part of the formula.

From one to all four

The standard question: one value, one quadrant, find the rest.

STEP	USING
find the partner of sin or cos	$sin^2\alpha + cos^2\alpha = 1$
fix its sign	the quadrant
form $tan\ \alpha$	$\frac{\sin\alpha}{\cos\alpha}$
form $cot\ \alpha$	$\frac{1}{\tan\alpha}$

Example. $tan\ \alpha = \frac{4}{3}$, quadrant III. Then $\frac{1}{\cos^2\alpha} = 1 + \frac{16}{9} = \frac{25}{9}$, so $cos^2\alpha = \frac{9}{25}$ and, in quadrant III, $cos\ \alpha = -\frac{3}{5}$. Hence $sin\ \alpha = tan\ \alpha \cdot cos\ \alpha = -\frac{4}{5}$ and $cot\ \alpha = \frac{3}{4}$.

THE 3–4–5 TRIANGLE IN DISGUISE

Almost every textbook question of this kind uses $3, 4, 5$ or $5, 12, 13$. Recognising the triple lets you write the three magnitudes down at sight, and leaves the only real work — the signs — properly visible.

Expressing one by another

Sometimes the question asks not for a number but for a formula.

WANTED	IN TERMS OF	ANSWER
$cos\ \alpha$	$sin\ \alpha$	$\pm\sqrt{1 - \sin^2\alpha}$
$tan\ \alpha$	$cos\ \alpha$	$\pm\frac{\sqrt{1 - \cos^2\alpha}}{\cos\alpha}$
$sin\ \alpha$	$tan\ \alpha$	$\pm\frac{\tan\alpha}{\sqrt{1 + \tan^2\alpha}}$

The ± is not laziness — without a quadrant it is the honest answer, and both branches really occur.

A SIMPLIFICATION THAT LOSES A SIGN IS WRONG, NOT MERELY UNTIDY

$\sqrt{\sin^2\alpha} = |sin\ \alpha|$, not $sin\ \alpha$. In quadrant III the two differ by a minus sign, and a question that supplies a quadrant is usually testing exactly that.

02 Worked examples

EXAMPLE 1 **Given $\tan\alpha = \frac{4}{3}$ in quadrant III, find the other three.**

1. $\frac{1}{\cos^2\alpha} = 1 + \frac{16}{9} = \frac{25}{9}$

2. $\cos^2\alpha = \frac{9}{25} \Rightarrow \cos\alpha = -\frac{3}{5}$

   Quadrant III.

3. $\sin\alpha = \tan\alpha \cdot \cos\alpha = \frac{4}{3} \cdot \left(-\frac{3}{5}\right) = -\frac{4}{5}$

4. $\cot\alpha = \frac{3}{4}$

ANSWER

$\sin = -\frac{4}{5}$, $\cos = -\frac{3}{5}$, $\cot = \frac{3}{4}$

EXAMPLE 2 **Simplify** $\dfrac{1}{1+\tan^2\alpha} + \dfrac{1}{1+\cot^2\alpha}$.

1. $$\frac{1}{1+\tan^2\alpha} = \cos^2\alpha$$

   The second relation.

2. $$\frac{1}{1+\cot^2\alpha} = \sin^2\alpha$$

   The third.

3. $\cos^2\alpha + \sin^2\alpha$

4. $= 1$

   For every α where both exist.

ANSWER

1

EXAMPLE 3 **Given $\sin\alpha - \cos\alpha = \frac{1}{2}$, find $\sin\alpha\cos\alpha$.**

1. Square both sides.

2. $\sin^2\alpha - 2\sin\alpha\cos\alpha + \cos^2\alpha = \frac{1}{4}$

3. $1 - 2\sin\alpha\cos\alpha = \frac{1}{4}$

   The first relation.

4. $\sin\alpha\cos\alpha = \frac{3}{8}$

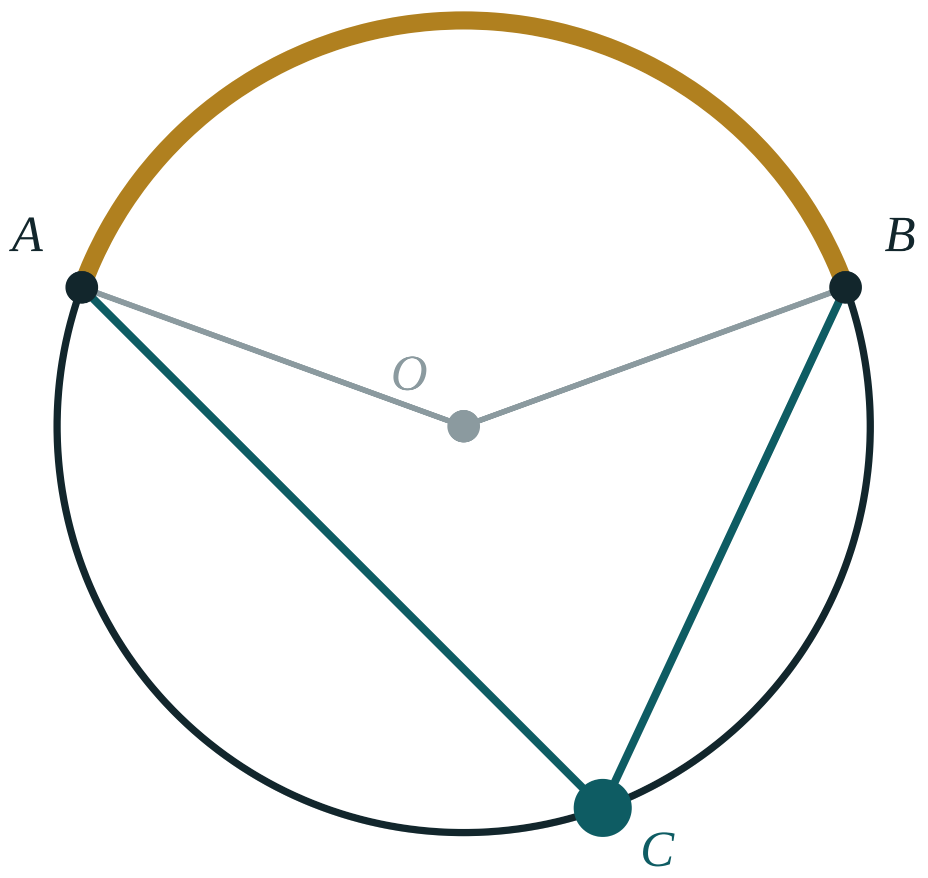
ANSWER

$\frac{3}{8}$

03 Interactive model

Derive the second relation on the board by dividing the first — and then rub it out and ask a pupil to do it again, so that nobody leaves believing it must be memorised.

One value fixes the rest



central ∠AOB **140°** | inscribed ∠ACB **70°** | ratio **2** | arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Relation	Munosabat	Соотношение
Identity	Ayniyat	Тождество
Reciprocal	Teskari son	Обратное число
Quotient	Bo‘linma	Частное
To divide through	Bo‘lib chiqish	Разделить почленно
Valid for	O‘rinli	Справедливо для
Restriction	Cheklov	Ограничение
Derive	Keltirib chiqarish	Вывести

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$1 + \tan^2\alpha$ equals:

- $\dfrac{1}{\sin^2\alpha}$
- $\dfrac{1}{\cos^2\alpha}$
- $\sin^2\alpha$
- $\cos^2\alpha$

$\tan\alpha \cdot \cot\alpha$ equals:

- 0
- 1
- $\sin\alpha$
- α

$1 + \cot^2\alpha$ is undefined when:

- $\cos\alpha = 0$
- $\sin\alpha = 0$
- $\alpha = 0$ only
- never

$\sqrt{\sin^2\alpha}$ equals:

- $\sin\alpha$
- $|\sin\alpha|$
- $-\sin\alpha$
- $\sin^2\alpha$

$\frac{1}{1+\tan^2\alpha}$ equals:

- $\sin^2\alpha$
- $\cos^2\alpha$
- $\tan^2\alpha$
- 1

To fix the sign of $\cos\alpha$ you need:

- the identity
- the quadrant
- a calculator
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\sin^2\alpha + \cos^2\alpha$

ANSWER 1

2. $\tan\alpha \cdot \cot\alpha$

ANSWER 1

3. $1 + \tan^2\alpha$

ANSWER $\dfrac{1}{\cos^2\alpha}$

4. $1 + \cot^2\alpha$

ANSWER $\dfrac{1}{\sin^2\alpha}$

5. $\tan\alpha$ in terms of $\sin, \cos$

ANSWER $\dfrac{\sin\alpha}{\cos\alpha}$

6. $\sin\alpha = \dfrac{3}{5}$, I: $\cos\alpha$

ANSWER $\dfrac{4}{5}$

7. $\cos\alpha = \dfrac{12}{13}$, IV: $\sin\alpha$

ANSWER $-\dfrac{5}{13}$

Medium · the standard question

7 PROBLEMS

1. $\tan\alpha = \dfrac{4}{3}$, III: $\cos\alpha$

   ANSWER $-\dfrac{3}{5}$

2. $\tan\alpha = \dfrac{4}{3}$, III: $\sin\alpha$

   ANSWER $-\dfrac{4}{5}$

3. $\cot\alpha = \dfrac{5}{12}$, I: $\sin\alpha$

   ANSWER $\dfrac{12}{13}$

4. Simplify $\sin^2\alpha + \sin^2\alpha \tan^2\alpha$

   ANSWER $\tan^2\alpha$

5. Simplify $\dfrac{1}{1 + \tan^2\alpha} + \dfrac{1}{1 + \cot^2\alpha}$

   ANSWER 1

6. Simplify $(1 + \tan^2\alpha) \cos^2\alpha$

   ANSWER 1

7. Simplify $\cot\alpha \cdot \sin\alpha$

   ANSWER $\cos\alpha$

Hard · stretch and reason

7 PROBLEMS

1. $\sin\alpha - \cos\alpha = \dfrac{1}{2}$: $\sin\alpha\cos\alpha$

ANSWER $\dfrac{3}{8}$

2. $\sin\alpha + \cos\alpha = \dfrac{7}{5}$: $\sin\alpha\cos\alpha$

ANSWER $\dfrac{12}{25}$

3. Simplify $\sin^4\alpha + \cos^4\alpha + 2\sin^2\alpha\cos^2\alpha$

ANSWER 1

4. Simplify $\dfrac{1 - \cos^2\alpha}{\sin\alpha\cos\alpha}$

ANSWER $\tan\alpha$

5. Prove $\tan^2\alpha - \sin^2\alpha = \tan^2\alpha\sin^2\alpha$

ANSWER Factor $\sin^2\alpha$ out of the left side

6. $\tan\alpha + \cot\alpha = 3$: $\sin\alpha\cos\alpha$

ANSWER $\dfrac{1}{3}$

7. Express $\sin\alpha$ by $\tan\alpha$

ANSWER $\pm\dfrac{\tan\alpha}{\sqrt{1 + \tan^2\alpha}}$

07 Homework

Homework — 5 tasks

Derive the second and third relations from the first at the top of the page before starting.

1. Given $\sin\alpha = -\dfrac{8}{17}$ in quadrant III, find the other three.

2. Given $cot\ \alpha = -\dfrac{3}{4}$ in quadrant II, find the other three.

3. Simplify $(1 + cot^2\alpha)\ sin^2\alpha$.

4. Given $sin\ \alpha + cos\ \alpha = \dfrac{1}{3}$, find $sin\ \alpha\ cos\ \alpha$.

5. Prove $cot^2\alpha - cos^2\alpha = cot^2\alpha\ cos^2\alpha$.

Proving trigonometric identities

Work on one side only, head for sines and cosines, and stop when the two sides agree.

LESSON 51–52 | 2 × 40 MIN | ALGEBRA 9, §22 | IGX 15.6

LESSON CLOCK

11' | 23' | 22' | 20' | 4'

What an identity asks	11 min
The default strategy	23 min
The standard moves	22 min
Permissible values	20 min
Homework	4 min

LEARNING OBJECTIVES

- Prove an identity by transforming one side into the other.
- Convert every function to sine and cosine as a default strategy.
- Use the difference of two squares and common denominators on trigonometric expressions.
- State the values of α for which an identity is not defined.

TEXTBOOK

National	pp. 120–125
Cambridge	Core & Extended, pp. 334–338
Workbook	Exercise 15.6

01 Explanation

What an identity asks

An **identity** is a statement that two expressions are equal for every permissible value of the variable. Proving one is not solving one.

	EQUATION	IDENTITY
true for	some α	all permissible α
you must	find them	transform
you may	operate on both sides	work on one side only
the answer is	a set	a chain of equalities

NEVER CROSS-MULTIPLY AN IDENTITY YOU HAVE NOT YET PROVED

> Starting from what you are trying to prove and manipulating both sides assumes the very thing in question. Take one side, and travel to the other; that argument is valid in one direction and cannot be criticised.

The default strategy

When nothing else suggests itself, do this:

1. Choose the more complicated side — there is more to simplify there.
2. Write every *tan* and *cot* as $\frac{\sin}{\cos}$ and $\frac{\cos}{\sin}$.
3. Put fractions over a common denominator.
4. Look for $\sin^2\alpha + \cos^2\alpha$ and replace it by 1.
5. Cancel, and compare with the other side.

Example. Prove $\tan\alpha + \cot\alpha = \dfrac{1}{\sin\alpha\cos\alpha}$.

$$\tan\alpha + \cot\alpha = \frac{\sin\alpha}{\cos\alpha} + \frac{\cos\alpha}{\sin\alpha} = \frac{\sin^2\alpha + \cos^2\alpha}{\sin\alpha\cos\alpha} = \frac{1}{\sin\alpha\cos\alpha}$$

SINES AND COSINES ARE THE COMMON LANGUAGE

Four functions can be combined in many ways; two can be combined in few. Converting everything to *sin* and *cos* removes the choices, and the proof usually falls out in two lines.

The standard moves

SEEING	DO	BECAUSE
$1 - \sin^2\alpha$	write $\cos^2\alpha$	the first relation
$1 - \cos^4\alpha$	factor as $(1 - \cos^2\alpha)(1 + \cos^2\alpha)$	difference of squares
$\dfrac{1}{1 - \sin\alpha}$	multiply by $\dfrac{1 + \sin\alpha}{1 + \sin\alpha}$	it creates $\cos^2\alpha$
two fractions	common denominator	the numerator often becomes 1
$\tan\alpha\sin\alpha + \cos\alpha$	convert and combine	it equals $\dfrac{1}{\cos\alpha}$

The third row is worth practising on its own: multiplying by the conjugate is exactly the rationalising trick from Chapter I, applied to a trigonometric denominator.

Permissible values

An identity is only claimed where both sides exist. Naming those values is part of the statement.

IDENTITY	EXCLUDED
$\tan\alpha + \cot\alpha = \dfrac{1}{\sin\alpha\cos\alpha}$	$\alpha = \dfrac{\pi k}{2}$
$\dfrac{1 - \cos^2\alpha}{\sin\alpha} = \sin\alpha$	$\alpha = \pi k$
$1 + \tan^2\alpha = \dfrac{1}{\cos^2\alpha}$	$\alpha = \dfrac{\pi}{2} + \pi k$

A CANCELLED DENOMINATOR DOES NOT REMOVE ITS RESTRICTION

$\dfrac{1 - \cos^2\alpha}{\sin\alpha}$ simplifies to $\sin\alpha$, which is defined everywhere — but the original expression is not, and the identity therefore still excludes $\alpha = \pi k$. The restriction belongs to the statement, not to the final form.

02 Worked examples

EXAMPLE 1 **Prove $\tan\alpha + \cot\alpha = \dfrac{1}{\sin\alpha\cos\alpha}$.**

1. Convert: $\dfrac{\sin\alpha}{\cos\alpha} + \dfrac{\cos\alpha}{\sin\alpha}$.

   Left side.

2. Common denominator $\sin\alpha\cos\alpha$.

3. Numerator $\sin^2\alpha + \cos^2\alpha = 1$.

4. $= \dfrac{1}{\sin\alpha\cos\alpha}$ — Q.E.D., for $\alpha \neq \dfrac{\pi k}{2}$.

ANSWER

Proved

EXAMPLE 2 **Prove $\dfrac{1}{1-\sin\alpha} + \dfrac{1}{1+\sin\alpha} = \dfrac{2}{\cos^2\alpha}$.**

1. Common denominator $(1-\sin\alpha)(1+\sin\alpha)$.

2. Numerator: $(1+\sin\alpha) + (1-\sin\alpha) = 2$.

3. Denominator: $1 - \sin^2\alpha = \cos^2\alpha$.

   Difference of squares.

4. $= \dfrac{2}{\cos^2\alpha}$ — Q.E.D., for $\cos\alpha \neq 0$.

ANSWER

Proved

EXAMPLE 3 **Prove $\sin^4\alpha - \cos^4\alpha = \sin^2\alpha - \cos^2\alpha$.**

1. Difference of two squares on the left.
2. $(\sin^2\alpha - \cos^2\alpha)(\sin^2\alpha + \cos^2\alpha)$
3. The second bracket is 1.
4. $= \sin^2\alpha - \cos^2\alpha$ — Q.E.D., for every α.

ANSWER

Proved

03 Interactive model

Prove one identity correctly and then “prove” it again by cross-multiplying from the statement itself; ask the class which argument they would accept, and why.

Test, then prove

$\sin^2 x + \cos^2 x$

x° **30** value **1**

Always 1, at every angle. Testing shows it; Pythagoras on the unit circle proves it.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Identity	Ayniyat	Тождество
Left-hand side	Chap tomon	Левая часть
Right-hand side	O‘ng tomon	Правая часть
Common denominator	Umumiy maxraj	Общий знаменатель
To transform	Almashtirish	Преобразовать
Permissible values	Joiz qiymatlar	Допустимые значения
Equivalent	Teng kuchli	Равносильный
Q.E.D.	Isbot tamom	Что и требовалось доказать

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

To prove an identity you work:

- on both sides
- on one side
- by cross-multiplying
- by testing values

The default first move is:

- square both sides
- convert to sin and cos
- differentiate
- substitute $\alpha = 0$

$1 - sin^2\alpha$ is:

- $cos^2\alpha$
- $tan^2\alpha$
- $cot^2\alpha$
- 1

$\tan\alpha + \cot\alpha$ equals:

- 1
- $\dfrac{1}{\sin\alpha\cos\alpha}$
- $\sin\alpha\cos\alpha$
- 2

$\frac{1}{1 - \sin\alpha}$ is multiplied by the conjugate to create:

- $\sin^2\alpha$
- $\cos^2\alpha$
- 1
- $\tan\alpha$

A cancelled denominator:

- removes its restriction
- keeps its restriction
- creates a new one
- is ignored

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Simplify $1 - \sin^2\alpha$

ANSWER $\cos^2\alpha$

2. Simplify $1 - \cos^2\alpha$

ANSWER $\sin^2\alpha$

3. Simplify $\cot\alpha \sin\alpha$

ANSWER $\cos\alpha$

4. Simplify $\tan\alpha \cos\alpha$

ANSWER $\sin\alpha$

5. Simplify $\sin^2\alpha + \cos^2\alpha + 1$

ANSWER 2

6. Simplify $\dfrac{\sin\alpha}{\cos\alpha} \cot\alpha$

ANSWER 1

7. Simplify $(1 - \sin\alpha)(1 + \sin\alpha)$

ANSWER $\cos^2\alpha$

Medium · the standard question

7 PROBLEMS

1. Prove $\tan\alpha + \cot\alpha = \dfrac{1}{\sin\alpha\cos\alpha}$

ANSWER Common denominator

2. Prove $\sin^4\alpha - \cos^4\alpha = \sin^2\alpha - \cos^2\alpha$

ANSWER Difference of squares

3. Prove $\dfrac{1}{1 - \sin\alpha} + \dfrac{1}{1 + \sin\alpha} = \dfrac{2}{\cos^2\alpha}$

ANSWER Conjugate denominator

4. Simplify $\tan\alpha \sin\alpha + \cos\alpha$

ANSWER $\dfrac{1}{\cos\alpha}$

5. Simplify $\dfrac{1 - \cos^2\alpha}{\sin\alpha\cos\alpha}$

ANSWER $\tan\alpha$

6. Prove $(1 + \tan^2\alpha)\cos^2\alpha = 1$

ANSWER The second relation

7. Simplify $\sin^2\alpha\cot^2\alpha + \sin^2\alpha$

ANSWER 1

Hard · stretch and reason

7 PROBLEMS

1. Prove $tan^2\alpha - sin^2\alpha = tan^2\alpha\ sin^2\alpha$

   ANSWER Factor $sin^2\alpha$

2. Prove $\dfrac{\cos\alpha}{1 - \sin\alpha} = \dfrac{1 + \sin\alpha}{\cos\alpha}$

   ANSWER Cross-check by the conjugate

3. Prove $sin^6\alpha + cos^6\alpha = 1 - 3\ sin^2\alpha\ cos^2\alpha$

   ANSWER Sum of cubes, then the first relation

4. Prove $\dfrac{1 + \cos\alpha}{\sin\alpha} + \dfrac{\sin\alpha}{1 + \cos\alpha} = \dfrac{2}{\sin\alpha}$

   ANSWER Common denominator

5. Simplify $(tan\ \alpha + cot\ \alpha)\ sin\ \alpha\ cos\ \alpha$

   ANSWER 1

6. Prove $\dfrac{\tan\alpha}{1 + \tan^2\alpha} = sin\ \alpha\ cos\ \alpha$

   ANSWER Use the second relation

7. Excluded values of $tan\ \alpha + cot\ \alpha = \dfrac{1}{\sin\alpha\cos\alpha}$

   ANSWER $\alpha = \dfrac{\pi k}{2}$

07 Homework

Homework — 5 tasks

End every proof with the values of α that had to be excluded.

1. Prove $(1 + cot^2\alpha)\ sin^2\alpha = 1$.
2. Prove $\dfrac{1}{1 - \cos\alpha} + \dfrac{1}{1 + \cos\alpha} = \dfrac{2}{\sin^2\alpha}$.

3. Prove $\cot^2\alpha - \cos^2\alpha = \cot^2\alpha \cos^2\alpha$.
4. Simplify $\cot\alpha \cos\alpha + \sin\alpha$.
5. Prove $\sin^4\alpha + \cos^4\alpha = 1 - 2\sin^2\alpha \cos^2\alpha$.

The functions of −α, and periodicity

Turning backwards reflects the point in the x-axis — and that one picture gives four formulae.

LESSON 53–54 | 2 × 40 MIN | ALGEBRA 9, §23 | IGX 15.7

LESSON CLOCK

13' | 20' | 23' | 20' | 4'

Turning the other way	13 min
Even and odd	20 min
Period	23 min
Reducing an angle	20 min
Homework	4 min

LEARNING OBJECTIVES

- Derive sin(−α) = −sin α and cos(−α) = cos α from the unit circle.
- Classify the four functions as even or odd.
- State the period of each function and use it to reduce an angle.
- Evaluate a function at a large or negative angle in one line.

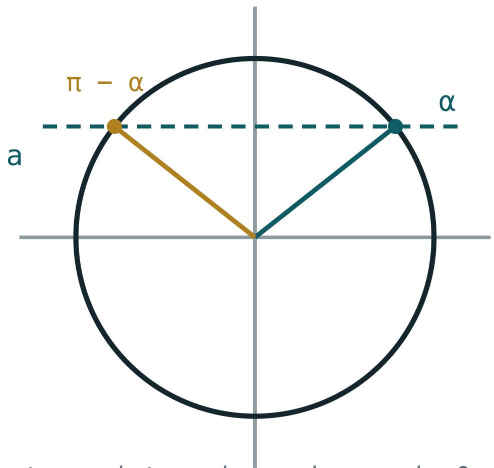
TEXTBOOK

National	pp. 126–130
Cambridge	Core & Extended, pp. 339–343
Workbook	Exercise 15.7

01 Explanation

Turning the other way

Turn $P(1, 0)$ through α to reach (x, y). Turning through $-\alpha$ — the same amount, clockwise — reaches $(x, -y)$: the mirror image in the x-axis.



α and −α land at points that are mirror images in the horizontal axis.

The abscissa is unchanged and the ordinate has changed sign, so:

> $\cos(-\alpha) = \cos\alpha \quad \sin(-\alpha) = -\sin\alpha$

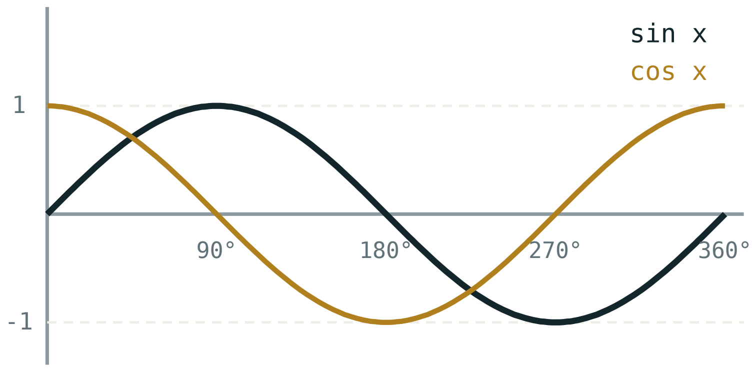
Dividing gives the other two: $\tan(-\alpha) = -\tan\alpha$ and $\cot(-\alpha) = -\cot\alpha$.

> ONE PICTURE, FOUR FORMULAE
>
> There is nothing here to memorise separately. Reflecting a point in the x-axis keeps x and flips y — and cosine **is** x, sine **is** y. Anyone who can draw the reflection can rebuild all four in seconds.

Even and odd

This is exactly the parity of Chapter I, applied to the new functions.

FUNCTION	$F(-A)$	PARITY	GRAPH SYMMETRIC ABOUT
$\cos\alpha$	$\cos\alpha$	even	the y-axis
$\sin\alpha$	$-\sin\alpha$	odd	the origin
$\tan\alpha$	$-\tan\alpha$	odd	the origin
$\cot\alpha$	$-\cot\alpha$	odd	the origin



period 360°, range −1 to 1

The cosine curve is symmetric in the vertical axis; the sine curve is symmetric about the origin.

Cosine is the only even one of the four, which is why it appears so often as the “safe” function in symmetric problems.

Period

A full turn returns the point to where it started, so adding 2π changes nothing:

$$\sin(\alpha + 2\pi k) = \sin\alpha \quad \cos(\alpha + 2\pi k) = \cos\alpha, \; k \in \mathbb{Z}$$

Tangent and cotangent repeat sooner. A turn of π sends (x, y) to $(-x, -y)$, and a quotient of two numbers is unchanged when both change sign:

$$\tan(\alpha + \pi k) = \tan\alpha \quad \cot(\alpha + \pi k) = \cot\alpha$$

FUNCTION	LEAST PERIOD
$\sin\alpha, \cos\alpha$	2π
$\tan\alpha, \cot\alpha$	π

4π IS A PERIOD OF SINE; IT IS NOT *THE* PERIOD

Any multiple of a period is again a period. “The period” always means the **least** positive one, and for sine that is 2π.

Reducing an angle

Parity and period together evaluate any angle at all.

EXPRESSION	STEP	VALUE
$\sin(-\frac{\pi}{6})$	odd	$-\frac{1}{2}$
$\cos(-\frac{\pi}{3})$	even	$\frac{1}{2}$
$\sin\frac{13\pi}{6}$	-2π	$\frac{1}{2}$
$\tan\frac{9\pi}{4}$	-2π	1
$\cos 750°$	$-720°$	$\frac{\sqrt{3}}{2}$

The order is always the same: strip whole periods first, then use parity if what remains is negative.

STRIP 2π FROM SINE AND COSINE, BUT π IS ENOUGH FOR TANGENT

$\tan\frac{9\pi}{4}$ can be reduced by 2π to $\tan\frac{\pi}{4}$, but also by π twice. Using the wrong period for the wrong function is the commonest error in this lesson.

02 Worked examples

EXAMPLE 1 **Evaluate $\sin(-\frac{\pi}{6}) + \cos(-\frac{\pi}{3})$.**

1. Sine is odd: $\sin(-\frac{\pi}{6}) = -\frac{1}{2}$.
2. Cosine is even: $\cos(-\frac{\pi}{3}) = \frac{1}{2}$.
3. $-\frac{1}{2} + \frac{1}{2}$
4. $= 0$

ANSWER

0

EXAMPLE 2 **Evaluate $\cos 750°$.**

1. $750 - 720 = 30$

   Two full turns.

2. $\cos 750° = \cos 30°$

3. Read the table.

4. $= \dfrac{\sqrt{3}}{2}$

ANSWER

$\dfrac{\sqrt{3}}{2}$

EXAMPLE 3 **Simplify $\dfrac{\sin(-\alpha) + \cos(-\alpha)}{\cos \alpha}$.**

1. Numerator: $-\sin \alpha + \cos \alpha$.

   Odd, then even.

2. $= \dfrac{\cos \alpha - \sin \alpha}{\cos \alpha}$

3. Split the fraction.

4. $= 1 - \tan \alpha$

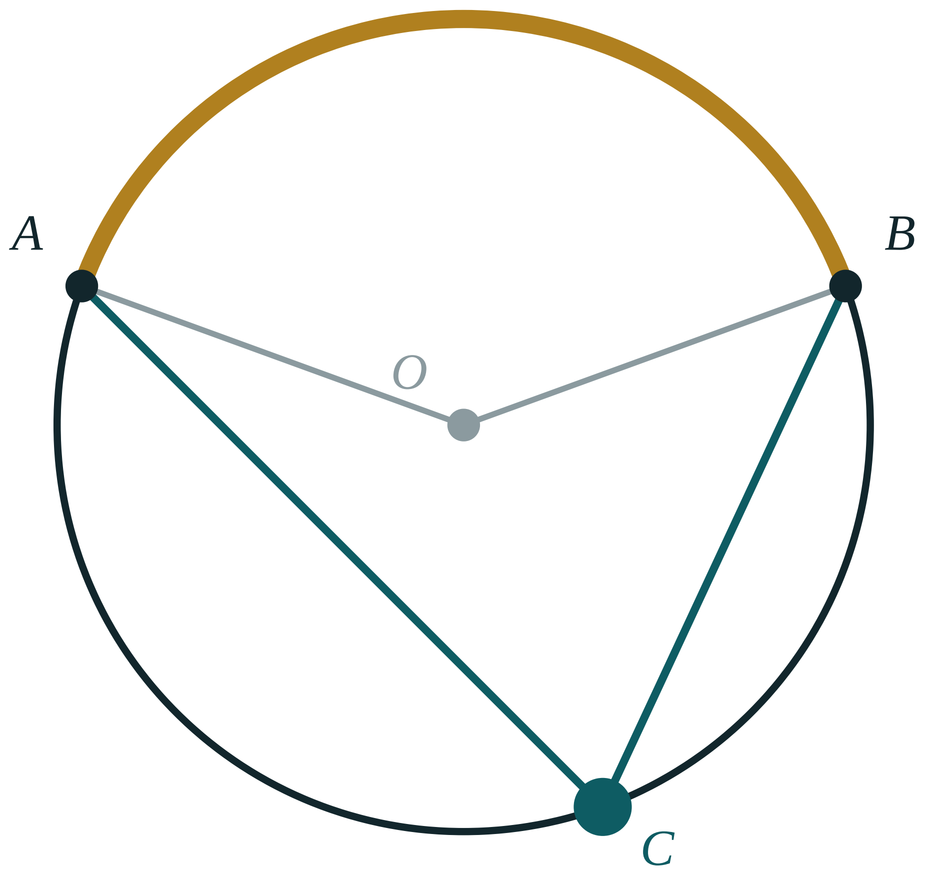
ANSWER

$1 - \tan \alpha$

03 Interactive model

Fold a sheet with the unit circle drawn on it along the x-axis; the two angles land on each other and the four parity formulae become one physical fact.

α and −α together



central ∠AOB **140°** | inscribed ∠ACB **70°** | ratio **2** | arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Even function	Juft funksiya	Чётная функция
Odd function	Toq funksiya	Нечётная функция
Reflection	Simmetriya	Отражение
Period	Davr	Период
Periodic	Davriy	Периодический
Least period	Eng kichik davr	Наименьший период
Reduce	Keltirish	Привести
Symmetric	Simmetrik	Симметричный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\cos(-\alpha)$ equals:

- $\cos\alpha$
- $-\cos\alpha$
- $\sin\alpha$
- $-\sin\alpha$

$\sin(-\alpha)$ equals:

- $\sin\alpha$
- $-\sin\alpha$
- $\cos\alpha$
- 0

The only even function of the four is:

- sin
- cos
- tan
- cot

The least period of $tan\ \alpha$:

- $\frac{\pi}{2}$
- π
- 2π
- 4π

$sin\ \underline{13\pi}\ 6$ equals:

- $\frac{1}{2}$
- $-\frac{1}{2}$
- $\frac{\sqrt{3}}{2}$
- 1

4π for sine is:

- the period
- a period
- not a period
- the least period

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $sin(-\frac{\pi}{6})$

   ANSWER $-\frac{1}{2}$

2. $cos(-\frac{\pi}{3})$

   ANSWER $\frac{1}{2}$

3. $tan(-\frac{\pi}{4})$

   ANSWER -1

4. $cos(-\pi)$

   ANSWER -1

5. $sin(-\frac{\pi}{2})$

   ANSWER -1

6. Period of $cos\ \alpha$

   ANSWER 2π

7. Period of $cot\ \alpha$

   ANSWER π

Medium · the standard question

7 PROBLEMS

1. $sin\ \frac{13\pi}{6}$

ANSWER $\frac{1}{2}$

2. $cos\ \frac{9\pi}{4}$

ANSWER $\frac{\sqrt{2}}{2}$

3. $tan\ \frac{9\pi}{4}$

ANSWER 1

4. $cos\ 750°$

ANSWER $\frac{\sqrt{3}}{2}$

5. $sin(-390°)$

ANSWER $-\frac{1}{2}$

6. Simplify $sin(-\alpha)\ cos(-\alpha)$

ANSWER $-sin\ \alpha\ cos\ \alpha$

7. Simplify $\frac{\sin(-\alpha) + \cos(-\alpha)}{\cos \alpha}$

ANSWER $1 - tan\ \alpha$

Hard · stretch and reason

7 PROBLEMS

1. Simplify $\sin(-\alpha) + \sin(\alpha + 2\pi)$

ANSWER 0

2. Simplify $\tan(-\alpha)\cot(\alpha + \pi)$

ANSWER -1

3. $\cos(-\frac{17\pi}{3})$

ANSWER $\frac{1}{2}$

4. $\sin(-1110°)$

ANSWER $-\frac{1}{2}$

5. Is $y = \sin\alpha\cos\alpha$ even or odd?

ANSWER Odd

6. Is $y = \sin^2\alpha$ even or odd?

ANSWER Even

7. Least period of $y = \sin 2\alpha$

ANSWER π

07 Homework

Homework — 5 tasks

Strip the whole periods first; only then use parity.

1. Evaluate $\sin(-\frac{\pi}{3}) + \cos(-\frac{\pi}{6})$.
2. Evaluate $\cos 840°$ and $\tan 405°$.
3. Simplify $\frac{\cos(-\alpha) - \sin(-\alpha)}{\sin\alpha}$.

4. Say whether $y = \cos\alpha \tan\alpha$ is even or odd, and prove it.

5. Give the least period of $y = \cos 3\alpha$.

The addition formulae for sine and cosine

The formulae that decide, once and for all, that sin(α + β) is not sin α + sin β.

LESSON 55–56 | 2 × 40 MIN | ALGEBRA 9, §24 | IGX 15.8

LESSON CLOCK

11' | 23' | 22' | 20' | 4'

A warning first	11 min
The four formulae	23 min
New exact values	22 min
Reading them backwards	20 min
Homework	4 min

LEARNING OBJECTIVES

- State the four addition formulae for sine and cosine accurately.
- Use them to find exact values at 15°, 75° and similar angles.
- Recognise the formulae read backwards, in order to compress an expression.
- Show by a counter-example that sine is not additive.

TEXTBOOK

National	pp. 131–137
Cambridge	Core & Extended, pp. 344–350
Workbook	Exercise 15.8

01 Explanation

A warning first

Before the formulae, the mistake they exist to prevent.

CLAIM	TEST AT $A = B = \frac{\pi}{2}$	VERDICT
$\sin(\alpha + \beta) = \sin\alpha + \sin\beta$	$\sin\pi = 0$	$1 + 1 = 2$ ✗
$\cos(\alpha + \beta) = \cos\alpha + \cos\beta$	$\cos\pi = -1$	$0 + 0 = 0$ ✗

SINE IS A FUNCTION, NOT A MULTIPLIER

sin written in front of a bracket is not a coefficient and cannot be distributed over the bracket. One substitution disposes of the idea permanently — and it is worth doing once, by hand, rather than being told.

The four formulae

$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

$$\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$$

$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$

$$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$$

PATTERN	SINE	COSINE
the products	mixed: sc and cs	alike: cc and ss
the sign	the same as in the bracket	the opposite

TWO SENTENCES INSTEAD OF FOUR FORMULAE

Sine mixes and keeps the sign; cosine matches and flips it. Learn those two sentences and all four formulae can be written out correctly, in either order, without hesitation.

New exact values

Any angle that is a sum or difference of two known angles now has an exact value.

Example. $15° = 45° - 30°$, so

$$\sin 15° = \sin 45° \cos 30° - \cos 45° \sin 30° = \frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2}\cdot\frac{1}{2} = \frac{\sqrt{6}-\sqrt{2}}{4}$$

ANGLE	AS	$\sin$	$\cos$
$15°$	$45° - 30°$	$\frac{\sqrt{6}-\sqrt{2}}{4}$	$\frac{\sqrt{6}+\sqrt{2}}{4}$
$75°$	$45° + 30°$	$\frac{\sqrt{6}+\sqrt{2}}{4}$	$\frac{\sqrt{6}-\sqrt{2}}{4}$

Notice that $\sin 75° = \cos 15°$: the two angles add to $90°$, and complementary angles always swap sine and cosine.

Reading them backwards

Half the examination questions on this topic use the formulae from right to left.

SEEING	WRITE
$\sin 40° \cos 20° + \cos 40° \sin 20°$	$\sin 60° = \dfrac{\sqrt{3}}{2}$
$\cos 70° \cos 10° + \sin 70° \sin 10°$	$\cos 60° = \dfrac{1}{2}$
$\sin 3\alpha \cos \alpha - \cos 3\alpha \sin \alpha$	$\sin 2\alpha$

LOOK FOR THE PATTERN BEFORE YOU COMPUTE

An expression with four trigonometric factors and one plus sign is almost never meant to be evaluated term by term. Match it against the four formulae first; the whole question usually collapses to a single known value.

02 Worked examples

EXAMPLE 1 **Find $\sin 15°$ exactly.**

1. $15° = 45° - 30°$
2. $\sin 45° \cos 30° - \cos 45° \sin 30°$

   Mix, keep the sign.
3. $\dfrac{\sqrt{2}}{2} \cdot \dfrac{\sqrt{3}}{2} - \dfrac{\sqrt{2}}{2} \cdot \dfrac{1}{2}$
4. $= \dfrac{\sqrt{6} - \sqrt{2}}{4}$

ANSWER

$$\frac{\sqrt{6} - \sqrt{2}}{4}$$

EXAMPLE 2 **Simplify $\sin 40° \cos 20° + \cos 40° \sin 20°$.**

1. Mixed products, plus sign — this is $\sin(\alpha + \beta)$.

2. $= \sin(40° + 20°)$

3. $= \sin 60°$

4. $= \dfrac{\sqrt{3}}{2}$

ANSWER

$$\frac{\sqrt{3}}{2}$$

EXAMPLE 3 **Given $\sin \alpha = \dfrac{3}{5}$ (I) and $\cos \beta = \dfrac{5}{13}$ (I), find $\sin(\alpha + \beta)$.**

1. $\cos \alpha = \dfrac{4}{5}$, $\sin \beta = \dfrac{12}{13}$.

   Both in quadrant I.

2. $\sin \alpha \cos \beta + \cos \alpha \sin \beta$

3. $\dfrac{3}{5} \cdot \dfrac{5}{13} + \dfrac{4}{5} \cdot \dfrac{12}{13}$

4. $= \dfrac{15}{65} + \dfrac{48}{65} = \dfrac{63}{65}$

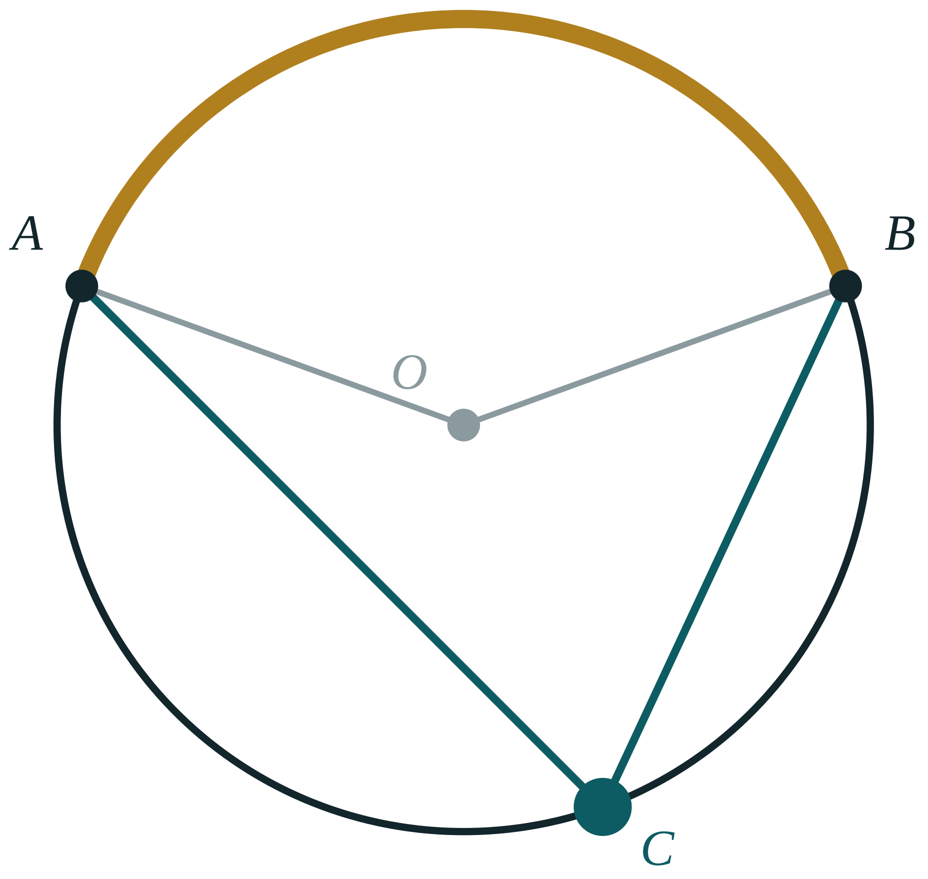
ANSWER

$$\frac{63}{65}$$

03 Interactive model

Open the lesson by asking the class to evaluate sin 90° two ways — directly, and as sin 45° + sin 45°; the contradiction motivates everything that follows.

Adding two angles



central ∠AOB **140°** | inscribed ∠ACB **70°** | ratio **2** | arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Addition formula	Qo‘shish formulasi	Формула сложения
Sum of angles	Burchaklar yig‘indisi	Сумма углов
Difference of angles	Burchaklar ayirmasi	Разность углов
Expand	Yoyish	Раскрыть
Compress	Yig‘ish	Свернуть
Counter-example	Qarama-qarshi misol	Контрпример
Exact value	Aniq qiymat	Точное значение
Substitution	O‘rniga qo‘yish	Подстановка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\sin(\alpha + \beta)$ equals:

- $\sin\alpha + \sin\beta$
- $\sin\alpha\cos\beta + \cos\alpha\sin\beta$
- $\cos\alpha\cos\beta - \sin\alpha\sin\beta$
- $2\sin\alpha\sin\beta$

$\cos(\alpha + \beta)$ equals:

- $\cos\alpha\cos\beta + \sin\alpha\sin\beta$
- $\cos\alpha\cos\beta - \sin\alpha\sin\beta$
- $\cos\alpha + \cos\beta$
- $\sin\alpha\cos\beta$

$\cos(\alpha - \beta)$ equals:

- $cc - ss$
- $cc + ss$
- $sc - cs$
- $sc + cs$

$\sin 40° \cos 20° + \cos 40° \sin 20°$ equals:

- $\sin 20°$
- $\sin 60°$
- $\cos 60°$
- $\sin 800°$

$\sin 15°$ equals:

- $\frac{\sqrt{6}-\sqrt{2}}{4}$
- $\frac{\sqrt{6}+\sqrt{2}}{4}$
- $\frac{1}{4}$
- $\frac{\sqrt{3}}{2}$

$\sin(\alpha + \beta) = \sin \alpha + \sin \beta$ is:

- always true
- never true
- true for some α, β
- an identity

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\sin(\alpha + \beta)$

   ANSWER $\sin\alpha\cos\beta + \cos\alpha\sin\beta$

2. $\cos(\alpha + \beta)$

   ANSWER $\cos\alpha\cos\beta - \sin\alpha\sin\beta$

3. $\sin(\alpha - \beta)$

   ANSWER $\sin\alpha\cos\beta - \cos\alpha\sin\beta$

4. $\cos(\alpha - \beta)$

   ANSWER $\cos\alpha\cos\beta + \sin\alpha\sin\beta$

5. $\sin 40^\circ \cos 20^\circ + \cos 40^\circ \sin 20^\circ$

   ANSWER $\frac{\sqrt{3}}{2}$

6. $\cos 70^\circ \cos 10^\circ + \sin 70^\circ \sin 10^\circ$

   ANSWER $\frac{1}{2}$

7. Is $\sin(\alpha + \beta) = \sin\alpha + \sin\beta$?

   ANSWER No

Medium · the standard question

7 PROBLEMS

1. $\sin 15°$

ANSWER $\dfrac{\sqrt{6}-\sqrt{2}}{4}$

2. $\cos 15°$

ANSWER $\dfrac{\sqrt{6}+\sqrt{2}}{4}$

3. $\sin 75°$

ANSWER $\dfrac{\sqrt{6}+\sqrt{2}}{4}$

4. $\cos 75°$

ANSWER $\dfrac{\sqrt{6}-\sqrt{2}}{4}$

5. Simplify $\sin 3\alpha \cos \alpha - \cos 3\alpha \sin \alpha$

ANSWER $\sin 2\alpha$

6. Simplify $\cos 5\alpha \cos 3\alpha + \sin 5\alpha \sin 3\alpha$

ANSWER $\cos 2\alpha$

7. $\sin \alpha = \dfrac{3}{5}$, $\cos \beta = \dfrac{5}{13}$, both I: $\sin(\alpha + \beta)$

ANSWER $\dfrac{63}{65}$

Hard · stretch and reason

7 PROBLEMS

1. $\sin\alpha = \dfrac{3}{5}$, $\cos\beta = \dfrac{5}{13}$, both I: $\cos(\alpha+\beta)$

   ANSWER $-\dfrac{16}{65}$

2. $\sin\alpha = \dfrac{4}{5}$ (II), $\cos\beta = \dfrac{3}{5}$ (I): $\sin(\alpha-\beta)$

   ANSWER $\dfrac{24}{25}$

3. Prove $\sin(\alpha+\beta)\sin(\alpha-\beta) = \sin^2\alpha - \sin^2\beta$

   ANSWER Expand both brackets

4. Simplify $\cos(60° + \alpha) + \cos(60° - \alpha)$

   ANSWER $\cos\alpha$

5. Simplify $\sin(45° + \alpha) - \sin(45° - \alpha)$

   ANSWER $\sqrt{2}\sin\alpha$

6. $\tan 15°$ from $\sin 15°$ and $\cos 15°$

   ANSWER $2-\sqrt{3}$

7. Prove $\cos(\alpha+\beta)\cos(\alpha-\beta) = \cos^2\alpha - \sin^2\beta$

   ANSWER Expand both brackets

07 Homework

Homework — 5 tasks

Write out the formula you are using before substituting; the mark is for the formula.

1. Find $\cos 15°$ and $\sin 75°$ exactly.
2. Simplify $\sin 50°\cos 10° - \cos 50°\sin 10°$.
3. Given $\sin\alpha = \dfrac{8}{17}$ (I) and $\cos\beta = \dfrac{4}{5}$ (I), find $\cos(\alpha-\beta)$.

4. Prove $\sin(\alpha + \beta) + \sin(\alpha - \beta) = 2 \sin \alpha \cos \beta$.

5. Show that $\cos(\alpha + \beta) = \cos \alpha + \cos \beta$ is false, with one substitution.

The tangent of a sum and of a difference

Divide one addition formula by the other and the tangent formula falls out.

LESSON 57–58 | 2 × 40 MIN | ALGEBRA 9, §25 | IGX 15.9

LESSON CLOCK

16' | 18' | 22' | 20' | 4'

The derivation	16 min
The conditions	18 min
Exact values	22 min
Backwards	20 min
Homework	4 min

LEARNING OBJECTIVES

- Derive tan(α ± β) from the sine and cosine addition formulae.
- State the values of α and β for which the formula is not defined.
- Find exact values such as tan 15° and tan 75°.
- Use the formula backwards to compress an expression.

TEXTBOOK

National	pp. 138–142
Cambridge	Core & Extended, pp. 351–355
Workbook	Exercise 15.9

01 Explanation

The derivation

Start from the definition and use the two formulae already proved:

$$\tan(\alpha+\beta) = \frac{\sin(\alpha+\beta)}{\cos(\alpha+\beta)} = \frac{\sin\alpha\cos\beta + \cos\alpha\sin\beta}{\cos\alpha\cos\beta - \sin\alpha\sin\beta}$$

Now divide numerator and denominator by $\cos\alpha\cos\beta$. Every term becomes a tangent:

$$\tan(\alpha+\beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$$

$$\tan(\alpha-\beta) = \frac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta}$$

THE SIGNS ARE CROSSED

A plus in the bracket puts a plus on top and a **minus** underneath. Getting this the wrong way round is the one error this formula is prone to, and checking it against $\tan(45° + 45°)$ — which must be undefined

— catches it instantly.

The conditions

The derivation divided by $\cos\alpha\cos\beta$, so the formula carries conditions the original sine and cosine formulae did not.

REQUIREMENT	WHY
$\tan\alpha, \tan\beta$	must exist: $\alpha, \beta \neq \frac{\pi}{2} + \pi k$
$\tan(\alpha+\beta)$	must exist: $\alpha + \beta \neq \frac{\pi}{2} + \pi k$
$1 - \tan\alpha\tan\beta \neq 0$	it is a denominator

The last two are the same condition seen from two sides: $\tan\alpha\tan\beta = 1$ happens exactly when $\alpha + \beta$ is an odd multiple of $\frac{\pi}{2}$.

$\tan 90°$ IS NOT "INFINITY"

It does not exist. When a calculation produces $1 - \tan\alpha\tan\beta = 0$ the correct statement is that $\tan(\alpha + \beta)$ is undefined — and often the intended answer is precisely that.

Exact values

ANGLE	AS	WORKING	VALUE
$\tan 15°$	$45° - 30°$	$\dfrac{1 - \frac{\sqrt{3}}{3}}{1 + \frac{\sqrt{3}}{3}}$	$2 - \sqrt{3}$
$\tan 75°$	$45° + 30°$	$\dfrac{1 + \frac{\sqrt{3}}{3}}{1 - \frac{\sqrt{3}}{3}}$	$2 + \sqrt{3}$
$\tan 105°$	$60° + 45°$	$\dfrac{\sqrt{3} + 1}{1 - \sqrt{3}}$	$-(2 + \sqrt{3})$

Each answer needs rationalising at the end — multiply above and below by the conjugate of the denominator, exactly as in Chapter I.

Notice that $\tan 15° \cdot \tan 75° = 1$: the angles are complementary, so the tangents are reciprocals.

Backwards

An expression of the shape $\dfrac{a + b}{1 - ab}$ where a and b are tangents is a compressed sum of angles.

SEEING	WRITE
$\dfrac{\tan 25° + \tan 20°}{1 - \tan 25° \tan 20°}$	$\tan 45° = 1$
$\dfrac{\tan 70° - \tan 10°}{1 + \tan 70° \tan 10°}$	$\tan 60° = \sqrt{3}$

THE ‘1’ IN THE DENOMINATOR IS THE CLUE

Nothing else in trigonometry produces a bare 1 added to or subtracted from a product of two tangents. When you see it, the whole expression is one tangent of one angle.

02 Worked examples

EXAMPLE 1 **Find *tan 15°* exactly.**

1. $15° = 45° - 30°$

2. $$\frac{\tan 45° - \tan 30°}{1 + \tan 45° \tan 30°} = \frac{1 - \frac{\sqrt{3}}{3}}{1 + \frac{\sqrt{3}}{3}}$$

3. Multiply above and below by 3: $\frac{3 - \sqrt{3}}{3 + \sqrt{3}}$.

4. Rationalise: $= 2 - \sqrt{3} \approx 0.268$.

ANSWER

$2 - \sqrt{3}$

EXAMPLE 2 **Simplify $\frac{\tan 25° + \tan 20°}{1 - \tan 25° \tan 20°}$.**

1. This is $\tan(\alpha + \beta)$ with $\alpha = 25°, \beta = 20°$.

2. $= \tan 45°$

3. $= 1$

4. No decimal work was needed at all.

ANSWER

1

EXAMPLE 3 **Given $\tan\alpha = 2$ and $\tan\beta = 3$, find $\tan(\alpha + \beta)$.**

1. $\dfrac{2 + 3}{1 - 2 \cdot 3}$
2. $= \dfrac{5}{-5}$
3. $= -1$
4. So $\alpha + \beta$ is an obtuse angle with tangent -1 — that is $135°$.

ANSWER

-1

03 Interactive model

Ask for tan(45° + 45°) from the formula; the denominator vanishes, and the class discovers the condition instead of being handed it.

Two tangents, one angle

$\tan(45° + x°)$

x° **15**

value **1.73**

Compare it with the addition formula $\dfrac{1 + \tan x°}{1 - \tan x°}$ — that is where this curve comes from.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Tangent of a sum	Yig‘indi tangensi	Тангенс суммы
Derivation	Keltirib chiqarish	Вывод
Denominator	Maxraj	Знаменатель
Divide through	Bo‘lib chiqish	Разделить почленно
Condition	Shart	Условие
Undefined	Aniqlanmagan	Не определён
Rationalise	Irratsionallikdan qutulish	Избавиться от иррациональности
Compress	Yig‘ish	Свернуть

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$tan(\alpha + \beta)$ equals:

- $\dfrac{\tan\alpha + \tan\beta}{1 + \tan\alpha\tan\beta}$
- $\dfrac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$
- $tan\ \alpha + tan\ \beta$
- $\dfrac{1}{\tan\alpha\tan\beta}$

$tan(\alpha - \beta)$ has denominator:

- $1 - tan\ \alpha\ tan\ \beta$
- $1 + tan\ \alpha\ tan\ \beta$
- $tan\ \alpha\ tan\ \beta$
- 1

$tan\ 15°$ equals:

- $2 - \sqrt{3}$
- $2 + \sqrt{3}$
- $\dfrac{\sqrt{3}}{3}$
- $\sqrt{3}$

$\frac{\tan 25^\circ + \tan 20^\circ}{1 - \tan 25^\circ \tan 20^\circ}$ equals:

- 0
- 1
- $\sqrt{3}$
- undefined

The formula fails when:

- $\tan\alpha \tan\beta = 0$
- $\tan\alpha \tan\beta = 1$
- $\alpha = \beta$
- never

$\tan 90^\circ$ is:

- ∞
- undefined
- 0
- 1

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $tan(\alpha + \beta)$

ANSWER $\dfrac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$

2. $tan(\alpha - \beta)$

ANSWER $\dfrac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta}$

3. $tan\ \alpha = 1, tan\ \beta = 1$: $tan(\alpha + \beta)$

ANSWER Undefined

4. $tan\ \alpha = 2, tan\ \beta = 3$: $tan(\alpha + \beta)$

ANSWER -1

5. $\dfrac{\tan 25° + \tan 20°}{1 - \tan 25° \tan 20°}$

ANSWER 1

6. $tan\ 45°$

ANSWER 1

7. $tan\ 60°$

ANSWER $\sqrt{3}$

Medium · the standard question

7 PROBLEMS

1. $\tan 15°$

ANSWER $2 - \sqrt{3}$

2. $\tan 75°$

ANSWER $2 + \sqrt{3}$

3. $\dfrac{\tan 70° - \tan 10°}{1 + \tan 70° \tan 10°}$

ANSWER $\sqrt{3}$

4. $\tan\alpha = \dfrac{1}{2}$, $\tan\beta = \dfrac{1}{3}$: $\tan(\alpha + \beta)$

ANSWER 1

5. $\tan\alpha = 3$, $\tan\beta = \dfrac{1}{2}$: $\tan(\alpha - \beta)$

ANSWER 1

6. $\tan 15° \cdot \tan 75°$

ANSWER 1

7. $\tan 105°$

ANSWER $-(2 + \sqrt{3})$

Hard · stretch and reason

7 PROBLEMS

1. Prove $\tan 15° \tan 75° = 1$

   ANSWER Complementary angles

2. $\tan\alpha = \dfrac{3}{4}$: $\tan(\alpha + 45°)$

   ANSWER 7

3. $\tan\alpha = \dfrac{3}{4}$: $\tan(\alpha - 45°)$

   ANSWER $-\dfrac{1}{7}$

4. Prove $\tan(\alpha + \beta)(1 - \tan\alpha \tan\beta) = \tan\alpha + \tan\beta$

   ANSWER Multiply the formula out

5. If $\alpha + \beta = 45°$, show $(1 + \tan\alpha)(1 + \tan\beta) = 2$

   ANSWER Expand and use the formula

6. $\tan\alpha = 1, \tan\beta = 2, \tan\gamma = 3$: show $\alpha + \beta + \gamma = \pi$

   ANSWER Add in two steps

7. Simplify $\dfrac{1 + \tan\alpha}{1 - \tan\alpha}$

   ANSWER $\tan(45° + \alpha)$

07 Homework

Homework — 5 tasks

Rationalise every surd answer; $\dfrac{1 - \dfrac{\sqrt{3}}{3}}{1 + \dfrac{\sqrt{3}}{3}}$ is not a finished answer.

1. Find $\tan 75°$ exactly.

2. Simplify $\dfrac{\tan 50° + \tan 10°}{1 - \tan 50° \tan 10°}$.

3. Given $tan\ \alpha = \dfrac{2}{3}$ and $tan\ \beta = \dfrac{1}{5}$, find $tan(\alpha + \beta)$.

4. Say for which α and β the formula for $tan(\alpha + \beta)$ cannot be used.

5. Show that $\dfrac{1 + \tan \alpha}{1 - \tan \alpha} = tan(45° + \alpha)$.

The double-angle formulae

Set β = α in the addition formulae and three of the most-used identities appear at once.

LESSON 59–60 | 2 × 40 MIN | ALGEBRA 9, §26 | IGX 15.10

LESSON CLOCK

13' | 21' | 22' | 20' | 4'

Setting β = α	13 min
Three faces of cos 2α	21 min
Lowering the degree	22 min
Both directions	20 min
Homework	4 min

LEARNING OBJECTIVES

- Derive sin 2α, cos 2α and tan 2α by putting β = α in the addition formulae.
- Give all three forms of cos 2α and choose the useful one.
- Use the half-angle consequences $2\sin^2\alpha = 1 - \cos 2\alpha$ and $2\cos^2\alpha = 1 + \cos 2\alpha$.
- Apply the formulae in both directions.

TEXTBOOK

National	pp. 143–148
Cambridge	Core & Extended, pp. 356–361
Workbook	Exercise 15.10

01 Explanation

Setting β = α

Nothing new has to be proved. Put $\beta = \alpha$ in each addition formula.

$$\sin 2\alpha = 2\sin\alpha\cos\alpha$$

$$\cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

$$\tan 2\alpha = \frac{2\tan\alpha}{1 - \tan^2\alpha}$$

FROM	PUTTING $B = A$	GIVES
$\sin\alpha\cos\beta + \cos\alpha\sin\beta$	$\sin\alpha\cos\alpha + \cos\alpha\sin\alpha$	$2\sin\alpha\cos\alpha$
$\cos\alpha\cos\beta - \sin\alpha\sin\beta$	$\cos^2\alpha - \sin^2\alpha$	$\cos 2\alpha$
$\dfrac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$	$\dfrac{2\tan\alpha}{1 - \tan^2\alpha}$	$\tan 2\alpha$

$\mathit{SIN\ 2A}$ IS NOT $\mathit{2\ SIN\ A}$

Test it at $\alpha = \frac{\pi}{2}$: the left side is $\sin\pi = 0$, the right is 2. The 2 multiplies the angle, not the function.

Three faces of cos 2α

Using $\sin^2\alpha + \cos^2\alpha = 1$, the cosine formula can be written three ways — and the choice of which to use is most of the skill.

$$\cos 2\alpha = \cos^2\alpha - \sin^2\alpha = 2\cos^2\alpha - 1 = 1 - 2\sin^2\alpha$$

FORM	USE IT WHEN
$\cos^2\alpha - \sin^2\alpha$	both are known
$2\cos^2\alpha - 1$	only $\cos\alpha$ is known
$1 - 2\sin^2\alpha$	only $\sin\alpha$ is known

Example. $\sin\alpha = \frac{3}{5}$. Then $\cos 2\alpha = 1 - 2 \cdot \frac{9}{25} = \frac{7}{25}$ — and the quadrant of α was never needed, because only $\sin^2\alpha$ appears.

SOMETIMES THE SIGN QUESTION DISAPPEARS

Any expression involving only even powers of sine and cosine is blind to the quadrant. Spotting that saves the whole sign discussion, and it is why the third form is so often the right one.

Lowering the degree

Read the last two forms backwards and the squares become linear in $\cos 2\alpha$:

$$\sin^2\alpha = \frac{1 - \cos 2\alpha}{2} \quad \cos^2\alpha = \frac{1 + \cos 2\alpha}{2}$$

These are the **degree-lowering** formulae, and they matter far beyond Grade 9 — every integral of $\sin^2 x$ in Grade 11 begins with them.

EXPRESSION	BECOMES
$\sin^2 15°$	$\frac{1 - \cos 30°}{2} = \frac{2 - \sqrt{3}}{4}$
$\cos^2 \frac{\pi}{8}$	$\frac{1 + \cos \frac{\pi}{4}}{2} = \frac{2 + \sqrt{2}}{4}$
$2\sin^2\alpha - 1$	$-\cos 2\alpha$

Both directions

SEEING	WRITE
$2\sin 15° \cos 15°$	$\sin 30° = \frac{1}{2}$
$\cos^2 22.5° - \sin^2 22.5°$	$\cos 45° = \frac{\sqrt{2}}{2}$
$\frac{2\tan 15°}{1 - \tan^2 15°}$	$\tan 30° = \frac{\sqrt{3}}{3}$
$\sin\alpha\cos\alpha$	$\frac{1}{2}\sin 2\alpha$

A PRODUCT OF A SINE AND A COSINE OF THE SAME ANGLE IS ALWAYS HALF A DOUBLE ANGLE

That single observation solves a large fraction of the questions in this section — including every “find the greatest value of $\sin\alpha\cos\alpha$” question, whose answer is $\frac{1}{2}$.

02 Worked examples

EXAMPLE 1 **Given $\sin\alpha = \frac{3}{5}$ in quadrant I, find $\sin 2\alpha$ and $\cos 2\alpha$.**

1. $\cos\alpha = \frac{4}{5}$

   Quadrant I.

2. $\sin 2\alpha = 2 \cdot \frac{3}{5} \cdot \frac{4}{5} = \frac{24}{25}$

3. $\cos 2\alpha = 1 - 2 \cdot \frac{9}{25}$

   Only $\sin\alpha$ needed.

4. $= \frac{7}{25}$

ANSWER

$\sin 2\alpha = \frac{24}{25}$, $\cos 2\alpha = \frac{7}{25}$

EXAMPLE 2 **Simplify $2 \sin 15° \cos 15°$ and $\cos^2 22.5° - \sin^2 22.5°$.**

1. The first is $\sin(2 \cdot 15°)$.
2. $= \sin 30° = \dfrac{1}{2}$
3. The second is $\cos(2 \cdot 22.5°)$.
4. $= \cos 45° = \dfrac{\sqrt{2}}{2}$

ANSWER

$\dfrac{1}{2}$ and $\dfrac{\sqrt{2}}{2}$

EXAMPLE 3 **Prove $\dfrac{\sin 2\alpha}{1 + \cos 2\alpha} = \tan \alpha$.**

1. Numerator: $2 \sin \alpha \cos \alpha$.
2. Denominator: $1 + 2\cos^2\alpha - 1 = 2\cos^2\alpha$.

   The second form.
3. $= \dfrac{2 \sin \alpha \cos \alpha}{2\cos^2\alpha}$
4. $= \tan \alpha$, for $\cos \alpha \neq 0$.

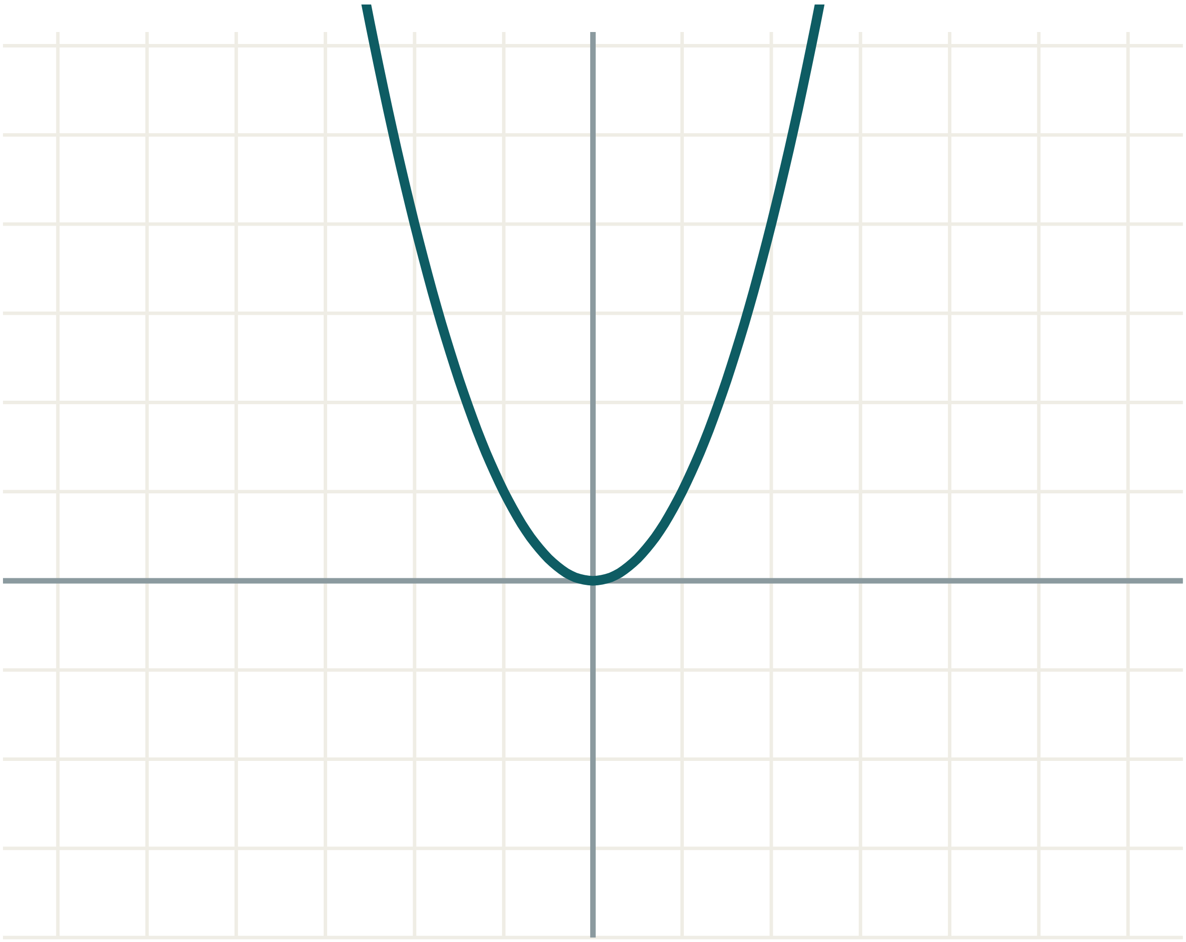
ANSWER

Proved

03 Interactive model

Ask the class to evaluate sin 2α at α = 90° both by the formula and by “2 sin α”; the contradiction settles the commonest misconception in one minute.

One angle, twice



a **1** | b **0** | c **0** | equation `y = f(x)`

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Double angle	Ikkilangan burchak	Двойной угол
Half angle	Yarim burchak	Половинный угол
Degree lowering	Darajani pasaytirish	Понижение степени
Equivalent form	Teng kuchli ko‘rinish	Равносильная форма
Substitute	O‘rniga qo‘ymoq	Подставить
Consequence	Natija	Следствие
Triple angle	Uchlangan burchak	Тройной угол
Verify	Tekshirmoq	Проверить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\sin 2\alpha$ equals:

- $2 \sin \alpha$
- $2 \sin \alpha \cos \alpha$
- $\sin^2\alpha$
- $\sin \alpha + \cos \alpha$

$\cos 2\alpha$ equals:

- $\cos^2\alpha + \sin^2\alpha$
- $\cos^2\alpha - \sin^2\alpha$
- $2 \cos \alpha$
- 1

Given only $\sin \alpha$, use:

- $\cos^2\alpha - \sin^2\alpha$
- $1 - 2\sin^2\alpha$
- $2\cos^2\alpha - 1$
- none

$sin^2\alpha$ equals:

- $\frac{1 - \cos 2\alpha}{2}$
- $\frac{1 + \cos 2\alpha}{2}$
- $\frac{\cos 2\alpha}{2}$
- $1 - cos\ 2\alpha$

$2\ sin\ 15°\ cos\ 15°$ equals:

- $\frac{1}{2}$
- $\frac{\sqrt{3}}{2}$
- 1
- $\frac{\sqrt{2}}{2}$

$tan\ 2\alpha$ equals:

- $2\ tan\ \alpha$
- $\frac{2 \tan \alpha}{1 - \tan^2\alpha}$
- $\frac{2 \tan \alpha}{1 + \tan^2\alpha}$
- $tan^2\alpha$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\sin 2\alpha$

   ANSWER $2 \sin \alpha \cos \alpha$

2. $\cos 2\alpha$ (basic form)

   ANSWER $\cos^2\alpha - \sin^2\alpha$

3. $\tan 2\alpha$

   ANSWER $\dfrac{2 \tan \alpha}{1 - \tan^2\alpha}$

4. $2 \sin 15° \cos 15°$

   ANSWER $\dfrac{1}{2}$

5. $\cos^2 22.5° - \sin^2 22.5°$

   ANSWER $\dfrac{\sqrt{2}}{2}$

6. $\sin \alpha \cos \alpha$ as a double angle

   ANSWER $\dfrac{1}{2} \sin 2\alpha$

7. Is $\sin 2\alpha = 2 \sin \alpha$?

   ANSWER No

Medium · the standard question

7 PROBLEMS

1. $\sin\alpha = \frac{3}{5}$ (I): $\sin 2\alpha$

ANSWER $\frac{24}{25}$

2. Same: $\cos 2\alpha$

ANSWER $\frac{7}{25}$

3. $\cos\alpha = \frac{4}{5}$ (IV): $\sin 2\alpha$

ANSWER $-\frac{24}{25}$

4. $\tan\alpha = \frac{1}{3}$: $\tan 2\alpha$

ANSWER $\frac{3}{4}$

5. $\sin^2 15°$

ANSWER $\frac{2-\sqrt{3}}{4}$

6. $\cos^2\frac{\pi}{8}$

ANSWER $\frac{2+\sqrt{2}}{4}$

7. Simplify $1 - 2\sin^2\alpha$

ANSWER $\cos 2\alpha$

Hard · stretch and reason

7 PROBLEMS

1. Prove $\dfrac{\sin 2\alpha}{1 + \cos 2\alpha} = \tan \alpha$

   ANSWER Use $2\cos^2\alpha$

2. Prove $\dfrac{1 - \cos 2\alpha}{\sin 2\alpha} = \tan \alpha$

   ANSWER Use $2\sin^2\alpha$

3. Greatest value of $\sin \alpha \cos \alpha$

   ANSWER $\dfrac{1}{2}$

4. Prove $\sin 3\alpha = 3 \sin \alpha - 4 \sin^3\alpha$

   ANSWER Write $3\alpha = 2\alpha + \alpha$

5. $\sin \alpha + \cos \alpha = \dfrac{1}{2}$: $\sin 2\alpha$

   ANSWER $-\dfrac{3}{4}$

6. Simplify $\cos^4\alpha - \sin^4\alpha$

   ANSWER $\cos 2\alpha$

7. Prove $\cot \alpha - \tan \alpha = 2 \cot 2\alpha$

   ANSWER Common denominator, then $\sin 2\alpha$

07 Homework

Homework — 5 tasks

For every $\cos 2\alpha$ choose the form that avoids finding a second function.

1. Given $\cos \alpha = \dfrac{5}{13}$ in quadrant IV, find $\sin 2\alpha$ and $\cos 2\alpha$.
2. Simplify $2 \sin 22.5° \cos 22.5°$ and $1 - 2 \sin^2 15°$.

3. Evaluate $\cos^2\frac{\pi}{12}$ by lowering the degree.

4. Prove $\frac{1-\cos 2\alpha}{\sin 2\alpha} = \tan\alpha$.

5. Given $\sin\alpha - \cos\alpha = \frac{1}{3}$, find $\sin 2\alpha$.

The reduction formulae

Every angle brought back to the first quadrant by two questions: does the name change, and what is the sign?

LESSON 61–62 | 2 × 40 MIN | ALGEBRA 9, §27 | IGX 15.11

LESSON CLOCK

13' | 21' | 22' | 20' | 4'

The two questions	13 min
Working the rule	21 min
The table it produces	22 min
Evaluating anything	20 min
Homework	4 min

LEARNING OBJECTIVES

- Reduce sin, cos, tan and cot of $\pi/2 \pm \alpha$, $\pi \pm \alpha$, $3\pi/2 \pm \alpha$ and $2\pi - \alpha$ to a first-quadrant function.
- Use the two-question rule instead of memorising a table of sixteen entries.
- Evaluate any trigonometric function of any angle exactly.
- Simplify expressions containing several shifted angles.

TEXTBOOK

National	pp. 149–154
Cambridge	Core & Extended, pp. 362–367
Workbook	Exercise 15.11

01 Explanation

The two questions

An expression such as $sin(\pi - \alpha)$ or $cos(\frac{3\pi}{2} + \alpha)$ can always be rewritten as ± a function of α alone. Two questions decide it.

1. **Does the name change?** Yes if the shift is $\frac{\pi}{2}$ or $\frac{3\pi}{2}$ — an **odd** multiple of $\frac{\pi}{2}$, measured from the vertical axis. No if it is π or 2π.
2. **What is the sign?** Treat α as a small acute angle, find the quadrant of the whole angle, and take the sign the **original** function has there.

SIXTEEN FORMULAE, OR TWO QUESTIONS

Textbooks print a table of sixteen entries and pupils memorise it for a week. The two questions above reproduce every entry in about five seconds, cannot be half-remembered, and still work in Grade 11. Learn the questions, not the table.

Working the rule

Example 1. $\sin(\pi - \alpha)$. The shift is π — horizontal — so the name stays $\sin$. With α small, $\pi - \alpha$ is in quadrant II, where sine is positive. Hence $\sin(\pi - \alpha) = \sin\alpha$.

Example 2. $\cos(\frac{\pi}{2} + \alpha)$. The shift is $\frac{\pi}{2}$ — vertical — so cosine becomes sine. With α small, $\frac{\pi}{2} + \alpha$ is in quadrant II, where **cosine** is negative. Hence $\cos(\frac{\pi}{2} + \alpha) = -\sin\alpha$.

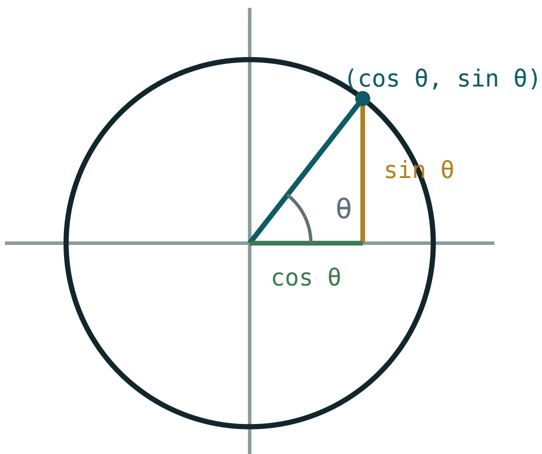
THE SIGN BELONGS TO THE ORIGINAL FUNCTION, NOT TO THE NEW ONE

In Example 2 the answer contains a sine, but the sign was decided by asking about **cosine** in quadrant II. Reversing that is the one mistake this rule is prone to.

The table it produces

	$\frac{\Pi}{2} - A$	$\frac{\Pi}{2} + A$	$\Pi - A$	$\Pi + A$	$\frac{3\Pi}{2} + A$	$2\Pi - A$
$\sin$	$\cos\alpha$	$\cos\alpha$	$\sin\alpha$	$-\sin\alpha$	$-\cos\alpha$	$-\sin\alpha$
$\cos$	$\sin\alpha$	$-\sin\alpha$	$-\cos\alpha$	$-\cos\alpha$	$\sin\alpha$	$\cos\alpha$
$\tan$	$\cot\alpha$	$-\cot\alpha$	$-\tan\alpha$	$\tan\alpha$	$-\cot\alpha$	$-\tan\alpha$
$\cot$	$\tan\alpha$	$-\tan\alpha$	$-\cot\alpha$	$\cot\alpha$	$-\tan\alpha$	$-\cot\alpha$

The first column is the familiar complementary-angle rule from Grade 8: $\sin(90° - \alpha) = \cos\alpha$. Everything else in the table is the same idea, moved round the circle.



on the unit circle the coordinates are cos θ and sin θ

Each shift moves the point to a new quadrant; the name and the sign follow from where it lands.

Evaluating anything

Combined with periodicity and parity, the reduction formulae evaluate every angle exactly.

ANGLE	STEP 1 — PERIOD	STEP 2 — REDUCE	VALUE
$\sin 150°$	—	$\sin(180° - 30°)$	$\frac{1}{2}$
$\cos 210°$	—	$\cos(180° + 30°)$	$-\frac{\sqrt{3}}{2}$
$\tan 300°$	—	$\tan(360° - 60°)$	$-\sqrt{3}$
$\sin 840°$	$-720° \to 120°$	$\sin(180° - 60°)$	$\frac{\sqrt{3}}{2}$
$\cos(-405°)$	even $\to 405° \to 45°$	—	$\frac{\sqrt{2}}{2}$

THE ORDER THAT ALWAYS WORKS

Strip whole turns, then use parity if the angle is negative, then reduce to the first quadrant, then read the exact-value table. Four steps, in that order, and no angle can resist them.

02 Worked examples

EXAMPLE 1 **Simplify** $\cos\left(\frac{\pi}{2} + \alpha\right)$.

1. Shift $\frac{\pi}{2}$ — vertical, so the name changes to sine.

   Question 1.

2. Take α small: the angle is in quadrant II.

   Question 2.

3. Cosine is negative in quadrant II.

   The original function.

4. $= -\sin\alpha$

ANSWER

$-\sin\alpha$

EXAMPLE 2 **Evaluate *sin 840°*.**

1. $840 - 720 = 120$

   Two full turns.

2. $\sin 120° = \sin(180° - 60°)$

3. Name stays; quadrant II; sine positive.

4. $= \sin 60° = \dfrac{\sqrt{3}}{2}$

ANSWER

$\dfrac{\sqrt{3}}{2}$

EXAMPLE 3 **Simplify** $\dfrac{\sin(\pi - \alpha)\cos\left(\dfrac{\pi}{2} + \alpha\right)}{\tan(\pi + \alpha)}$.

1. $\sin(\pi - \alpha) = \sin \alpha$

2. $\cos\left(\dfrac{\pi}{2} + \alpha\right) = -\sin \alpha$

3. $\tan(\pi + \alpha) = \tan \alpha$

4. $= \dfrac{\sin \alpha \cdot (-\sin \alpha)}{\tan \alpha} = -\sin \alpha \cos \alpha$

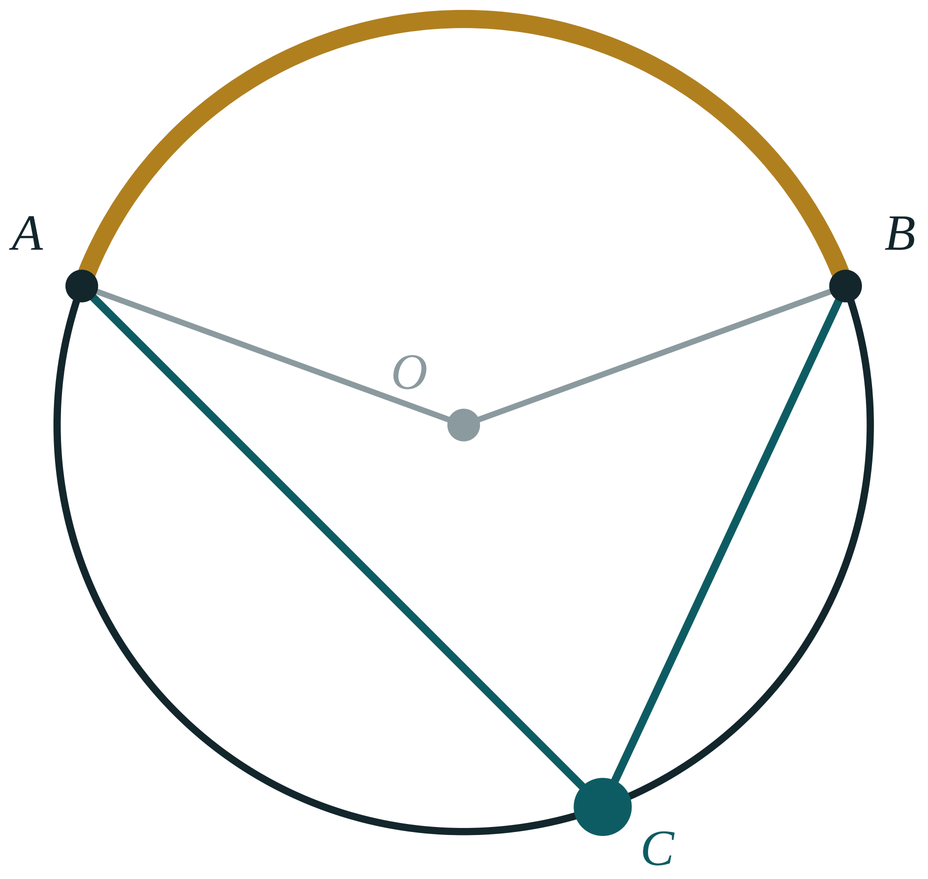
ANSWER

$-\sin \alpha \cos \alpha$

03 Interactive model

Do not hand out the table. Give six expressions and have the class produce the table themselves by asking the two questions; the table they write is the one they will remember.

Shift the angle



central ∠AOB **140°** | inscribed ∠ACB **70°** | ratio **2** | arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Reduction formula	Keltirish formulasi	Формула приведения
Co-function	Ko-funksiya	Кофункция
Shift	Siljish	Сдвиг
Vertical axis	Vertikal o‘q	Вертикальная ось
Horizontal axis	Gorizontal o‘q	Горизонтальная ось
Acute angle	O‘tkir burchak	Острый угол
Name changes	Nomi o‘zgaradi	Название меняется
Rule	Qoida	Правило

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The name changes for a shift of:

- π
- $\frac{\pi}{2}$
- 2π
- never

$\sin(\pi - \alpha)$ equals:

- $\sin \alpha$
- $-\sin \alpha$
- $\cos \alpha$
- $-\cos \alpha$

$\cos(\frac{\pi}{2} + \alpha)$ equals:

- $\sin \alpha$
- $-\sin \alpha$
- $\cos \alpha$
- $-\cos \alpha$

$\tan(\pi + \alpha)$ equals:

- $\tan\alpha$
- $-\tan\alpha$
- $\cot\alpha$
- $-\cot\alpha$

$\cos 210°$ equals:

- $\frac{\sqrt{3}}{2}$
- $-\frac{\sqrt{3}}{2}$
- $\frac{1}{2}$
- $-\frac{1}{2}$

The sign is decided by:

- the new function
- the original function
- the quadrant of α
- the period

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\sin(\pi - \alpha)$

   ANSWER $\sin\alpha$

2. $\cos(\pi - \alpha)$

   ANSWER $-\cos\alpha$

3. $\sin(\pi + \alpha)$

   ANSWER $-\sin\alpha$

4. $\cos(2\pi - \alpha)$

   ANSWER $\cos\alpha$

5. $\sin(\frac{\pi}{2} - \alpha)$

   ANSWER $\cos\alpha$

6. $\tan(\pi + \alpha)$

   ANSWER $\tan\alpha$

7. $\cos(\frac{\pi}{2} - \alpha)$

   ANSWER $\sin\alpha$

Medium · the standard question

7 PROBLEMS

1. $cos(\frac{\pi}{2} + \alpha)$

ANSWER $-sin\ \alpha$

2. $tan(\frac{\pi}{2} - \alpha)$

ANSWER $cot\ \alpha$

3. $cot(\frac{3\pi}{2} + \alpha)$

ANSWER $-tan\ \alpha$

4. $sin\ 150°$

ANSWER $\frac{1}{2}$

5. $cos\ 210°$

ANSWER $-\frac{\sqrt{3}}{2}$

6. $tan\ 300°$

ANSWER $-\sqrt{3}$

7. $sin\ 840°$

ANSWER $\frac{\sqrt{3}}{2}$

Hard · stretch and reason

7 PROBLEMS

1. Simplify $\dfrac{\sin(\pi-\alpha)\cos(\frac{\pi}{2}+\alpha)}{\tan(\pi+\alpha)}$

ANSWER $-\sin\alpha\cos\alpha$

2. Simplify $\sin(\frac{\pi}{2}+\alpha)+\cos(\pi+\alpha)$

ANSWER 0

3. Simplify $\tan(\pi-\alpha)\cot(\frac{\pi}{2}+\alpha)$

ANSWER $\tan^2\alpha$

4. $\cos(-405°)$

ANSWER $\dfrac{\sqrt{2}}{2}$

5. $\sin 1230°$

ANSWER $\dfrac{1}{2}$

6. Simplify $\sin^2(\pi-\alpha)+\cos^2(\frac{\pi}{2}+\alpha)$

ANSWER $2\sin^2\alpha$

7. Simplify $\dfrac{\cos(2\pi-\alpha)}{\sin(\frac{\pi}{2}-\alpha)}$

ANSWER 1

07 Homework

Homework — 5 tasks

Write the two questions and their answers beside every reduction; the working is the mark.

1. Simplify $\sin(\frac{3\pi}{2} - \alpha)$ and $\cos(\frac{3\pi}{2} - \alpha)$.

2. Evaluate $\sin 225°$, $\cos 315°$ and $\tan 240°$.

3. Evaluate $\cos 1110°$.

4. Simplify $\dfrac{\sin(\pi + \alpha)\cos(2\pi - \alpha)}{\sin(\frac{\pi}{2} + \alpha)}$.

5. Prove $\sin(\pi - \alpha) + \sin(\pi + \alpha) = 0$.

Sums and differences of sines and cosines

Turning a sum into a product — the step that makes a trigonometric equation factorisable.

LESSON 63–64 | 2 × 40 MIN | ALGEBRA 9, §28 | IGX 15.12

LESSON CLOCK

13' | 20' | 23' | 20' | 4'

Why a product is better	13 min
The four formulae	20 min
The derivation	23 min
Using them	20 min
Homework	4 min

LEARNING OBJECTIVES

- State the four sum-to-product formulae.
- Derive them by adding and subtracting the addition formulae.
- Use them to factorise and hence to solve simple equations.
- Convert a product back into a sum when that is the simpler direction.

TEXTBOOK

National	pp. 155–160
Cambridge	Core & Extended, pp. 368–372
Workbook	Exercise 15.12

01 Explanation

Why a product is better

An equation such as $\sin 3x + \sin x = 0$ cannot be attacked directly: there is no rule that lets a sum of two sines be zero term by term. Written as a product it becomes trivial — a product is zero exactly when one factor is.

FORM	WHAT YOU CAN DO
a sum	very little
a product $= 0$	set each factor to zero
a product $\neq 0$	estimate, bound, differentiate

THE SAME MOVE AS IN CHAPTER I

Factorising to solve is exactly what was done with quadratics and with the method of intervals. Trigonometry does not introduce a new strategy here; it introduces the tool that makes the old strategy available.

The four formulae

$$\sin\alpha + \sin\beta = 2\sin\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}$$

$$\sin\alpha - \sin\beta = 2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$$

$$\cos\alpha + \cos\beta = 2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}$$

$$\cos\alpha - \cos\beta = -2\sin\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$$

NOTICE	DETAIL
every right side	has the factor 2 and the half-sum and half-difference
the sine formulae	are mixed: one sine, one cosine
the cosine formulae	are alike: both cosines, or both sines
the last one	carries a minus sign — the only one that does

THE MINUS IN $\cos A - \cos B$ IS NOT OPTIONAL

It is the single most-forgotten sign in the whole of Grade 9 trigonometry. A useful check: $\cos 0 - \cos\pi = 2$, and the formula gives $-2\sin\frac{\pi}{2}\sin(-\frac{\pi}{2}) = 2$ ✓.

The derivation

Write $\alpha = u + v$ and $\beta = u - v$, so that $u = \frac{\alpha+\beta}{2}$ and $v = \frac{\alpha-\beta}{2}$. Then

$$\sin\alpha + \sin\beta = \sin(u+v) + \sin(u-v)$$

Expanding both by the addition formulae, the $\cos u \sin v$ terms cancel and the $\sin u \cos v$ terms double:

$$= 2 \sin u \cos v = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

The other three come out the same way, by subtracting instead of adding, or by starting from the cosine formulae.

Using them

Example. Solve $\sin 3x + \sin x = 0$.

STEP	WORKING
sum to product	$2 \sin 2x \cos x = 0$
first factor	$\sin 2x = 0 \Rightarrow x = \frac{\pi k}{2}$
second factor	$\cos x = 0 \Rightarrow x = \frac{\pi}{2} + \pi k$
combine	$x = \frac{\pi k}{2}$

The second family is contained in the first, so one expression covers everything.

The reverse direction is also useful: $2 \sin \alpha \cos \beta = \sin(\alpha + \beta) + \sin(\alpha - \beta)$ turns a product into a sum, which is what an integral in Grade 11 will want.

WHICH DIRECTION?

Going to a product helps when you must **solve**; going to a sum helps when you must **estimate or integrate**. Deciding which of the two you need before starting saves most of the work.

02 Worked examples

EXAMPLE 1 **Simplify $\sin 75° + \sin 15°$.**

1. Half sum $45°$, half difference $30°$.
2. $2 \sin 45° \cos 30°$
3. $2 \cdot \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2}$
4. $= \frac{\sqrt{6}}{2}$

ANSWER

$\frac{\sqrt{6}}{2}$

EXAMPLE 2 **Solve $\sin 3x + \sin x = 0$.**

1. $2 \sin 2x \cos x = 0$

   Half sum $2x$, half difference x.

2. $\sin 2x = 0 \Rightarrow x = \dfrac{\pi k}{2}$

3. $\cos x = 0 \Rightarrow x = \dfrac{\pi}{2} + \pi k$

   Already included.

4. $x = \dfrac{\pi k}{2}, k \in \mathbb{Z}$

ANSWER

$x = \dfrac{\pi k}{2}$

EXAMPLE 3 **Prove $\dfrac{\sin\alpha + \sin\beta}{\cos\alpha + \cos\beta} = \tan\dfrac{\alpha+\beta}{2}$.**

1. Numerator: $2 \sin \dfrac{\alpha+\beta}{2} \cos \dfrac{\alpha-\beta}{2}$.

2. Denominator: $2 \cos \dfrac{\alpha+\beta}{2} \cos \dfrac{\alpha-\beta}{2}$.

3. The common factors cancel.

4. $= \tan \dfrac{\alpha+\beta}{2}$

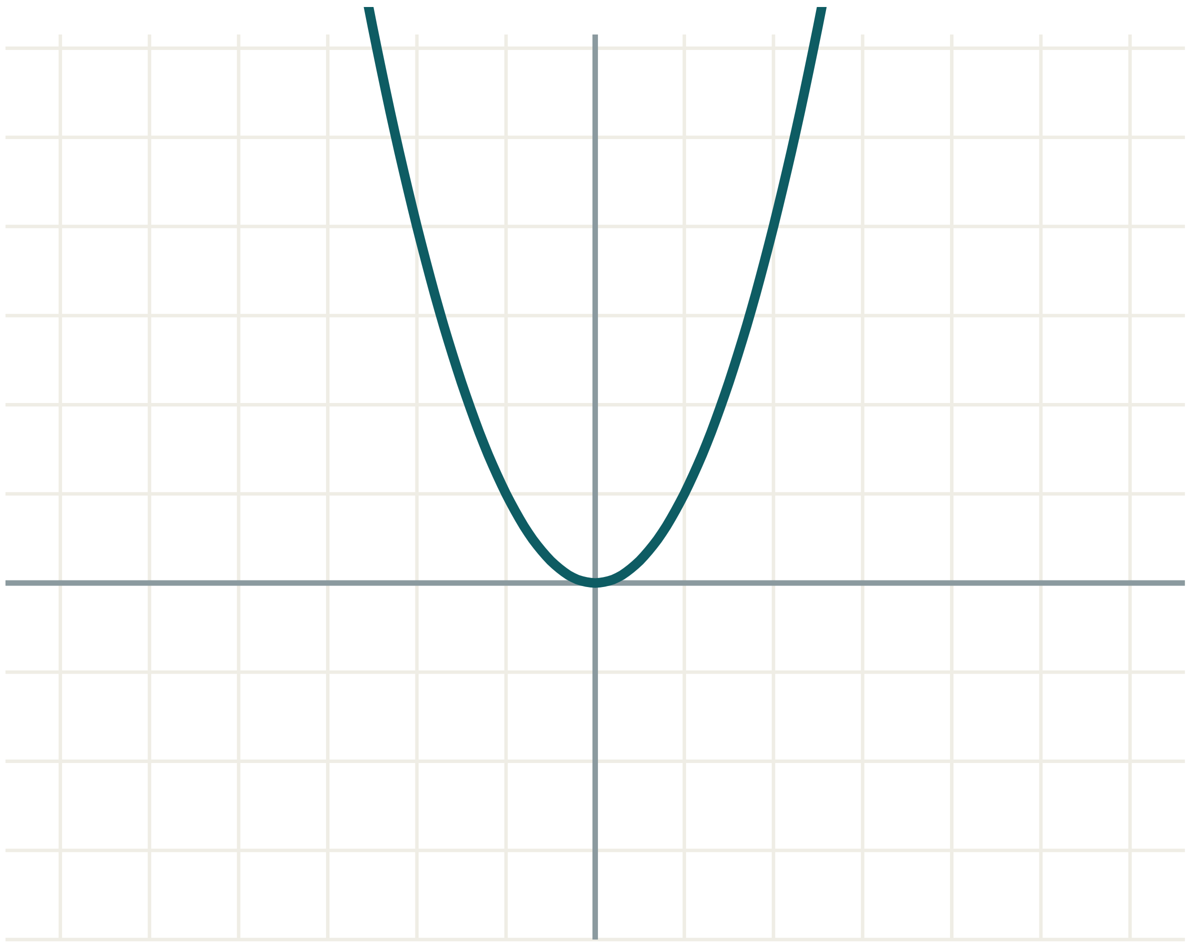
ANSWER

Proved

03 Interactive model

Check the minus sign in cos α − cos β on the board with α = 0, β = π; a formula verified once in front of the class is remembered with its sign.

A sum of two waves



a **1** | b **0** | c **0** | equation `y = f(x)`

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	OʻZBEKCHA	РУССКИЙ
Sum to product	Yigʻindini koʻpaytmaga	Сумма в произведение
Half sum	Yarim yigʻindi	Полусумма
Half difference	Yarim ayirma	Полуразность
Factorise	Koʻpaytuvchilarga ajratish	Разложить на множители
Product	Koʻpaytma	Произведение
Coefficient	Koeffitsiyent	Коэффициент
Auxiliary angle	Yordamchi burchak	Вспомогательный угол
Zero product	Nol koʻpaytma	Нулевое произведение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\sin\alpha + \sin\beta$ equals:

- $2\sin\dfrac{\alpha+\beta}{2}\cos\dfrac{\alpha-\beta}{2}$
- $2\cos\dfrac{\alpha+\beta}{2}\sin\dfrac{\alpha-\beta}{2}$
- $\sin(\alpha+\beta)$
- $2\sin\alpha\sin\beta$

$\cos\alpha - \cos\beta$ equals:

- $2\sin\dfrac{\alpha+\beta}{2}\sin\dfrac{\alpha-\beta}{2}$
- $-2\sin\dfrac{\alpha+\beta}{2}\sin\dfrac{\alpha-\beta}{2}$
- $2\cos\dfrac{\alpha+\beta}{2}\cos\dfrac{\alpha-\beta}{2}$
- $\cos(\alpha-\beta)$

$sin\ 75° + sin\ 15°$ equals:

- $\frac{\sqrt{6}}{2}$
- $\frac{\sqrt{2}}{2}$
- 1
- $\frac{\sqrt{3}}{2}$

$sin\ 3x + sin\ x$ factorises to:

- $2\ sin\ 2x\ cos\ x$
- $2\ sin\ x\ cos\ 2x$
- $sin\ 4x$
- $2\ sin\ 4x$

A product is zero when:

- both factors are
- at least one factor is
- neither is
- always

$\dfrac{\sin\alpha + \sin\beta}{\cos\alpha + \cos\beta}$ equals:

- $\tan\dfrac{\alpha+\beta}{2}$
- $\tan\dfrac{\alpha-\beta}{2}$
- $\cot\dfrac{\alpha+\beta}{2}$
- 1

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up — 7 PROBLEMS

1. $\sin\alpha + \sin\beta$

ANSWER $2\sin\dfrac{\alpha+\beta}{2}\cos\dfrac{\alpha-\beta}{2}$

2. $\sin\alpha - \sin\beta$

ANSWER $2\cos\dfrac{\alpha+\beta}{2}\sin\dfrac{\alpha-\beta}{2}$

3. $\cos\alpha + \cos\beta$

ANSWER $2\cos\dfrac{\alpha+\beta}{2}\cos\dfrac{\alpha-\beta}{2}$

4. $\cos\alpha - \cos\beta$

ANSWER $-2\sin\dfrac{\alpha+\beta}{2}\sin\dfrac{\alpha-\beta}{2}$

5. Factorise $\sin 3x + \sin x$

ANSWER $2\sin 2x\cos x$

6. Factorise $\cos 3x + \cos x$

ANSWER $2\cos 2x\cos x$

7. $\sin 75° + \sin 15°$

ANSWER $\dfrac{\sqrt{6}}{2}$

Medium · the standard question

7 PROBLEMS

1. $\cos 75° + \cos 15°$

ANSWER $\dfrac{\sqrt{6}}{2}$

2. $\sin 75° - \sin 15°$

ANSWER $\dfrac{\sqrt{2}}{2}$

3. $\cos 15° - \cos 75°$

ANSWER $\dfrac{\sqrt{2}}{2}$

4. Factorise $\sin 5x - \sin x$

ANSWER $2 \cos 3x \sin 2x$

5. Factorise $\cos 5x - \cos x$

ANSWER $-2 \sin 3x \sin 2x$

6. Solve $\sin 3x + \sin x = 0$

ANSWER $x = \dfrac{\pi k}{2}$

7. Simplify $\dfrac{\sin \alpha + \sin \beta}{\cos \alpha + \cos \beta}$

ANSWER $\tan \dfrac{\alpha + \beta}{2}$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\cos 3x - \cos x = 0$

ANSWER $x = \frac{\pi k}{2}$

2. Simplify $\frac{\sin\alpha - \sin\beta}{\cos\alpha + \cos\beta}$

ANSWER $\tan\frac{\alpha - \beta}{2}$

3. Simplify $\frac{\cos\alpha - \cos\beta}{\sin\alpha + \sin\beta}$

ANSWER $-\tan\frac{\alpha - \beta}{2}$

4. Prove $\sin\alpha + \sin 3\alpha + \sin 5\alpha = \sin 3\alpha(1 + 2\cos 2\alpha)$

ANSWER Pair the outer two first

5. Solve $\sin 5x = \sin x$

ANSWER $x = \frac{\pi k}{2}$ or $x = \frac{\pi}{6} + \frac{\pi k}{3}$

6. Write $2\sin 3\alpha\cos\alpha$ as a sum

ANSWER $\sin 4\alpha + \sin 2\alpha$

7. Greatest value of $\sin\alpha + \sin(60° - \alpha)$

ANSWER 1, at $\alpha = 30°$

07 Homework

Homework — 5 tasks

Write the half-sum and the half-difference explicitly before applying any formula.

1. Simplify $\cos 75° + \cos 15°$ and $\sin 105° - \sin 15°$.
2. Factorise $\sin 7x + \sin 3x$ and $\cos 7x - \cos 3x$.

3. Solve $cos\ 3x + cos\ x = 0$.

4. Prove $\dfrac{\sin\alpha - \sin\beta}{\cos\alpha + \cos\beta} = tan\ \dfrac{\alpha - \beta}{2}$.

5. Write $2\ cos\ 5\alpha\ cos\ 3\alpha$ as a sum.

Chapter exercises — the graphs of the trigonometric functions

Cambridge draws the three curves and reads the answers off them; the algebra and the picture must agree.

LESSON 65–66 | 2 × 40 MIN | ALGEBRA 9, III BOB MASHQLARI | IGX 15.13–15.14

LESSON CLOCK

16' | 20' | 22' | 18' | 4'

The three curves	16 min
What to read off	20 min
Solutions from a graph	22 min
Stretching the curve	18 min
Homework	4 min

LEARNING OBJECTIVES

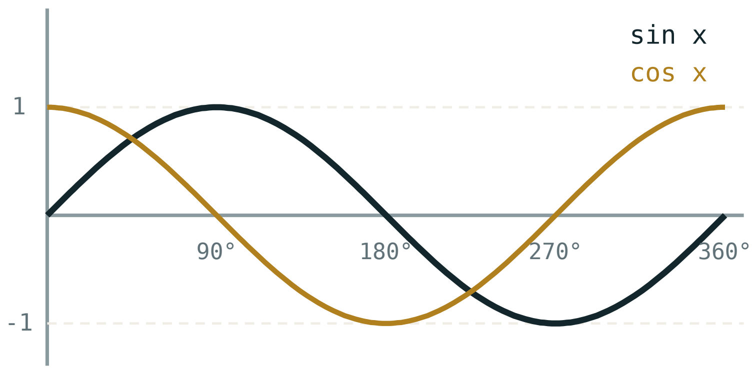
- Sketch y = sin x, y = cos x and y = tan x over 0° ≤ x ≤ 360°.
- Read amplitude, period and asymptotes from a sketch.
- Find all solutions of a simple equation in a given interval from the graph.
- Describe the effect of y = a sin bx on the curve.

TEXTBOOK

National	pp. 161–166
Cambridge	Core & Extended, pp. 373–382
Workbook	Exercise 15.13

01 Explanation

The three curves



period 360°, range −1 to 1

y = sin x and y = cos x — the same wave, a quarter turn apart.

	Y = SIN X	Y = COS X	Y = TAN X
period	$360°$	$360°$	$180°$
range	$[-1, 1]$	$[-1, 1]$	$\mathbb{R}$
at $x = 0$	0	1	0
zeros	$180°k$	$90° + 180°k$	$180°k$
asymptotes	none	none	$x = 90° + 180°k$
symmetry	odd	even	odd

THE COSINE CURVE IS THE SINE CURVE, SHIFTED

$\cos x = \sin(x + 90°)$ — which is one of the reduction formulae from the last lesson, read as a statement about graphs. Anyone who can draw one curve can draw the other by sliding it $90°$ to the left.

What to read off

A Cambridge question rarely says “find the period”; it says “write down the coordinates of the maximum point” or “state the value of k”. Both mean the same reading.

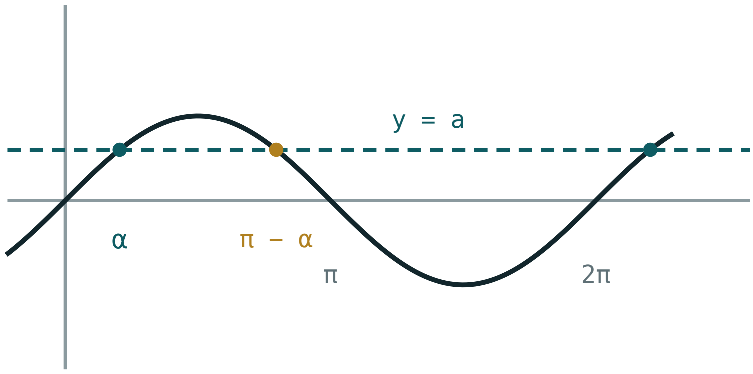
ASKED FOR	READ FROM THE GRAPH
amplitude	half the vertical distance between the highest and lowest points
period	the horizontal distance between two consecutive maxima
maximum point	coordinates, not just the height
number of solutions	count the crossings of the horizontal line

“AMPLITUDE” IS A HEIGHT, “MAXIMUM POINT” IS A PAIR OF NUMBERS

For $y = 3 \sin x$ the amplitude is 3 and the first maximum point is $(90°, 3)$. Answering with the wrong kind of object loses the mark even when the number is right.

Solutions from a graph

To solve $\sin x = 0.5$ for $0° \le x \le 360°$, draw the line $y = 0.5$ across the sine curve and count the crossings.



$$x = \alpha + 2\pi k \text{ and } x = \pi - \alpha + 2\pi k$$

One horizontal line, two crossings — a calculator gives only the first.

EQUATION	CROSSINGS IN $[0°, 360°]$	SOLUTIONS
$\sin x = 0.5$	2	$30°, 150°$
$\cos x = 0.5$	2	$60°, 300°$
$\tan x = 1$	2	$45°, 225°$
$\sin x = -0.5$	2	$210°, 330°$
$\sin x = 1$	1	$90°$
$\sin x = 2$	0	none

THE CALCULATOR GIVES ONE SOLUTION; THE GRAPH GIVES ALL OF THEM

$\sin^{-1} 0.5 = 30°$ and the machine stops there. The second solution, $150°$, exists because the curve comes back down — and a question that asks for “all solutions in the interval” is testing exactly whether you drew the picture.

Stretching the curve

This is the graph transformation work of Chapter I applied to a wave.

CURVE	AMPLITUDE	PERIOD	EFFECT
$y = \sin x$	1	$360°$	—
$y = 3\sin x$	3	$360°$	three times as tall
$y = \sin 2x$	1	$180°$	twice as fast
$y = 2\cos 3x$	2	$120°$	both at once
$y = \sin x + 2$	1	$360°$	lifted by 2

In general $y = a\sin bx$ has amplitude $|a|$ and period $\frac{360°}{|b|}$. The coefficient outside stretches vertically; the coefficient inside squeezes horizontally — the reverse of what most learners first expect.

02 Worked examples

EXAMPLE 1 **Solve $\sin x = 0.5$ for $0° \leq x \leq 360°$.**

1. $\sin^{-1} 0.5 = 30°$

   The calculator value.

2. Sine is positive in quadrants I and II.

3. The second solution is $180° - 30° = 150°$.

4. $x = 30°$ or $x = 150°$.

ANSWER

$30°, 150°$

EXAMPLE 2 **State the amplitude and period of $y = 4 \sin 3x$, and its first maximum point.**

1. Amplitude $= |4| = 4$.
   The outside coefficient.

2. Period $= \dfrac{360°}{3} = 120°$.
   The inside coefficient.

3. The first maximum is a quarter of a period in: $x = 30°$.

4. Maximum point $(30°, 4)$.

ANSWER

Amplitude 4, period $120°$, maximum $(30°, 4)$

EXAMPLE 3 **How many solutions has $\cos x = 0.3$ for $0° \le x \le 720°$?**

1. The period is $360°$.

2. In one period the line $y = 0.3$ crosses twice.

3. The interval is two periods long.

4. 4 solutions.

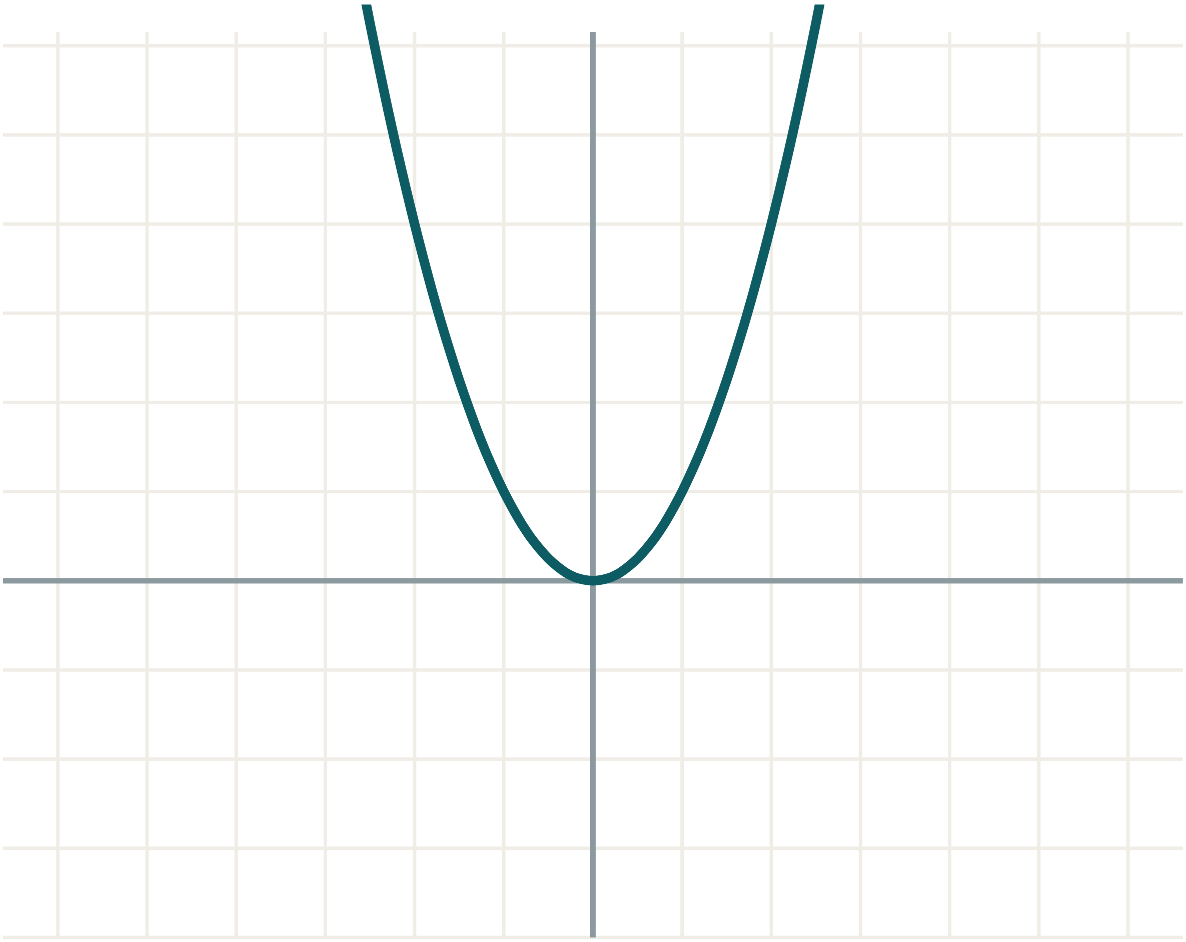
ANSWER

4

03 Interactive model

Ask for all solutions of sin x = 0.5 before mentioning the graph; let the class produce only 30°, then draw the curve and let them find what they missed.

a sin bx



a **1**

b **0**

c **0**

equation `y = f(x)`

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Amplitude	Amplituda	Амплитуда
Period	Davr	Период
Asymptote	Asimptota	Асимптота
Maximum	Maksimum	Максимум
Minimum	Minimum	Минимум
Symmetry of the curve	Egri chiziq simmetriyasi	Симметрия кривой
Interval	Oraliq	Промежуток
Sketch	Chizma	Эскиз

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The period of $y = tan\ x$:

- $90°$
- $180°$
- $360°$
- $720°$

The amplitude of $y = 3\ sin\ x$:

- 1
- 3
- 6
- $360°$

The period of $y = sin\ 2x$:

- $90°$
- $180°$
- $360°$
- $720°$

$\sin x = 0.5$ in $[0°, 360°]$ has:

- one solution
- two solutions
- three
- none

$\sin x = 2$ has:

- one solution
- two
- no solution
- infinitely many

$y = \tan x$ has asymptotes at:

- $180°k$
- $90° + 180°k$
- $360°k$
- none

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Period of $y = sin\ x$

   ANSWER $360°$

2. Period of $y = tan\ x$

   ANSWER $180°$

3. Range of $y = cos\ x$

   ANSWER $[-1, 1]$

4. $y = cos\ x$ at $x = 0$

   ANSWER 1

5. Amplitude of $y = 5\ sin\ x$

   ANSWER 5

6. Period of $y = sin\ 3x$

   ANSWER $120°$

7. Is $y = cos\ x$ even or odd?

   ANSWER Even

Medium · the standard question

7 PROBLEMS

1. Solve $\sin x = 0.5$ in $[0°, 360°]$

ANSWER $30°, 150°$

2. Solve $\cos x = 0.5$ in $[0°, 360°]$

ANSWER $60°, 300°$

3. Solve $\tan x = 1$ in $[0°, 360°]$

ANSWER $45°, 225°$

4. Solve $\sin x = -0.5$ in $[0°, 360°]$

ANSWER $210°, 330°$

5. Amplitude and period of $y = 4 \sin 3x$

ANSWER 4 and $120°$

6. First maximum point of $y = 4 \sin 3x$

ANSWER $(30°, 4)$

7. Solutions of $\cos x = 0.3$ in $[0°, 720°]$

ANSWER 4

Hard · stretch and reason

7 PROBLEMS

1. Solve $\cos x = -1$ in $[0°, 720°]$

ANSWER $180°, 540°$

2. Solve $2 \sin x + 1 = 0$ in $[0°, 360°]$

ANSWER $210°, 330°$

3. Solve $\sin 2x = 0.5$ in $[0°, 360°]$

ANSWER $15°, 75°, 195°, 255°$

4. Range of $y = 2 \cos x + 1$

ANSWER $[-1, 3]$

5. Least value of $y = 3 - 2 \sin x$

ANSWER 1

6. Number of solutions of $\tan x = 2$ in $[0°, 720°]$

ANSWER 4

7. Write $y = \cos x$ as a shifted sine curve

ANSWER $y = \sin(x + 90°)$

07 Homework

Homework — 5 tasks

Sketch the curve for every equation, even when you are sure of the answer.

1. Sketch $y = \sin x$ and $y = \cos x$ for $0° \le x \le 360°$ on one pair of axes.
2. Solve $\cos x = -0.5$ for $0° \le x \le 360°$.
3. State the amplitude and period of $y = 2 \sin 4x$.
4. How many solutions has $\sin x = 0.8$ for $0° \le x \le 1080°$?
5. Give the range of $y = 3 \cos x - 1$.

Home / Grade 9 / Algebra / Quarter 3

Control work 5, and the trigonometry chapter closed

Identities, addition, double angles and reduction — the whole of Chapter III in one paper.

LESSON 67–68 | 2 × 40 MIN | ALGEBRA 9, NAZORAT ISHI 5 | IGX 15 REVIEW

LESSON CLOCK

3 | 40' | 12' | 20' | 5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Use the fundamental relations and a quadrant to find all four functions.
- Apply an addition or double-angle formula accurately under time.
- Reduce an angle to the first quadrant and evaluate it exactly.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 114–166
Cambridge	Core & Extended, pp. 328–382
Workbook	Control paper A5

01 Explanation

The paper — 30 marks, 40 minutes

Q	TASK	MARKS	FROM
1	$\cos\alpha = -\dfrac{3}{5}$, quadrant II: find $\sin\alpha$ and $\tan\alpha$	5	L49–50
2	Prove $\tan\alpha + \cot\alpha = \dfrac{1}{\sin\alpha\cos\alpha}$	5	L51–52
3	Find $\cos 15°$ exactly	5	L55–56
4	$\sin\alpha = \dfrac{5}{13}$ (I): find $\sin 2\alpha$ and $\cos 2\alpha$	6	L59–60
5	Evaluate $\sin 225°$ and $\cos 750°$	5	L61–62
6	Factorise $\sin 5x - \sin x$	4	L63–64

WHERE THE MARKS ACTUALLY GO

Q1 carries two marks for the sign; Q2 one for stating the excluded values; Q3 one for rationalising; Q4 two for choosing the form of $\cos 2\alpha$ that needs only $\sin\alpha$; Q5 one for stripping the two whole turns before reducing.

Naming the slip

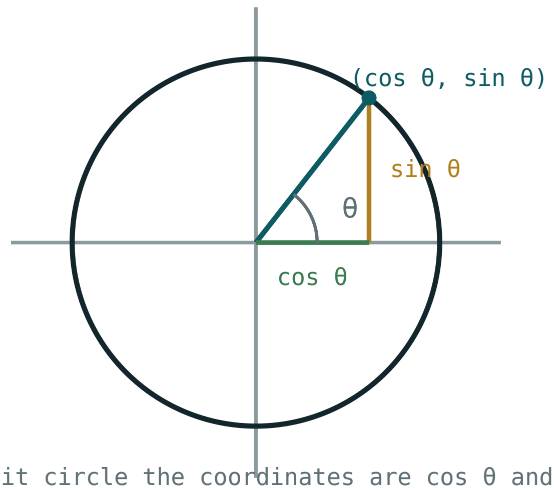
SLIP	WHAT IT LOOKS LIKE	THE FIX
sign left as ±	$\sin\alpha = \pm\dfrac{4}{5}$	quadrant II ⇒ positive
identity proved both ways at once	cross-multiplying the statement	start on one side
addition formula signs crossed	$\cos 15° = cc + ss$ *with a minus*	cosine matches and flips
surd not rationalised	$\dfrac{1}{2+\sqrt{3}}$	multiply by the conjugate
wrong form of $\cos 2\alpha$	finding $\cos\alpha$ unnecessarily	$1 - 2\sin^2\alpha$
$\sin 2\alpha = 2\sin\alpha$	$\dfrac{10}{13}$	$2\sin\alpha\cos\alpha$
reduction sign from the new function	$\cos(\dfrac{\pi}{2} + \alpha) = \sin\alpha$	ask about cosine, not sine

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

Chapter III as one map

Six boxes, and every arrow is a derivation the class has done:

- **the unit circle** → ***$\sin^2\alpha + \cos^2\alpha = 1$*** — “Pythagoras with hypotenuse 1”
- **that identity, divided** → ***$1 + \tan^2\alpha$*** and ***$1 + \cot^2\alpha$***
- **the addition formulae** → ***$\tan(\alpha \pm \beta)$*** — “divide one by the other”
- ***$\beta = \alpha$*** → **the double angle** — “and three faces of $\cos 2\alpha$”
- ***$\cos 2\alpha$* read backwards** → **degree lowering** — “$\sin^2\alpha = \dfrac{1 - \cos 2\alpha}{2}$”
- **adding two addition formulae** → **sum to product** — “and hence factorising”



on the unit circle the coordinates are cos θ and sin θ

One circle; every formula in the chapter is a statement about a point on it.

LOOKING FORWARD

Chapter IV changes subject entirely: sequences, arithmetic progressions and their sums. It needs nothing from trigonometry — but the habit of deriving rather than memorising, which this chapter was built to teach, is exactly what makes the progression formulae easy.

02 Worked examples

EXAMPLE 1 **Model answer, Q1: $\cos\alpha = -\dfrac{3}{5}$ in quadrant II.**

1. $\sin^2\alpha = 1 - \dfrac{9}{25} = \dfrac{16}{25}$

2. $\sin\alpha = \pm\dfrac{4}{5}$

   The size.

3. Quadrant II $\Rightarrow$ $\sin\alpha = \dfrac{4}{5}$.

   The sign.

4. $\tan\alpha = \dfrac{4/5}{-3/5} = -\dfrac{4}{3}$

ANSWER

$\sin\alpha = \dfrac{4}{5}$, $\tan\alpha = -\dfrac{4}{3}$

EXAMPLE 2 **Model answer, Q3: find *cos 15°*.**

1. $15° = 45° - 30°$

2. $\cos 45° \cos 30° + \sin 45° \sin 30°$

   Match, flip: minus becomes plus.

3. $$\frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2}$$

4. $$= \frac{\sqrt{6} + \sqrt{2}}{4}$$

ANSWER

$$\frac{\sqrt{6} + \sqrt{2}}{4}$$

EXAMPLE 3 **Model answer, Q4: $\sin\alpha = \frac{5}{13}$ in quadrant I.**

1. $\cos\alpha = \frac{12}{13}$

   Quadrant I.

2. $\sin 2\alpha = 2 \cdot \frac{5}{13} \cdot \frac{12}{13} = \frac{120}{169}$

3. $\cos 2\alpha = 1 - 2 \cdot \frac{25}{169}$

   Only $\sin\alpha$ needed.

4. $= \frac{119}{169}$

ANSWER

$\sin 2\alpha = \frac{120}{169}$, $\cos 2\alpha = \frac{119}{169}$

03 Interactive model

Put a wrong reduction on the board — cos(π/2 + α) = sin α — and let the class find the error by asking the two questions rather than by consulting a table.

Chapter III in twelve questions

$1 + \tan^2\alpha$ equals:

- $\dfrac{1}{\sin^2\alpha}$
- $\dfrac{1}{\cos^2\alpha}$
- $\sin^2\alpha$
- 1

To fix a sign you need:

- the identity
- the quadrant
- a calculator
- nothing

An identity is proved by working:

- on both sides
- on one side
- by testing
- by cross-multiplying

$cos(-\alpha)$ equals:

- $cos\ \alpha$
- $-cos\ \alpha$
- $sin\ \alpha$
- $-sin\ \alpha$

The least period of $tan\ \alpha$:

- $\frac{\pi}{2}$
- π
- 2π
- 4π

$sin(\alpha + \beta)$ equals:

- $sin\ \alpha + sin\ \beta$
- $sc + cs$
- $cc - ss$
- $2\ sin\ \alpha\ cos\ \beta$

$cos(\alpha - \beta)$ equals:

- $cc - ss$
- $cc + ss$
- $sc + cs$
- $sc - cs$

$tan(\alpha + \beta)$ has denominator:

- $1 + \tan\alpha \tan\beta$
- $1 - \tan\alpha \tan\beta$
- $\tan\alpha \tan\beta$
- 1

$\sin 2\alpha$ equals:

- $2 \sin\alpha$
- $2 \sin\alpha \cos\alpha$
- $\sin^2\alpha$
- $1 - \cos 2\alpha$

$sin^2\alpha$ equals:

- $\frac{1 - \cos 2\alpha}{2}$
- $\frac{1 + \cos 2\alpha}{2}$
- $\frac{\cos 2\alpha}{2}$
- $1 - \cos 2\alpha$

$\cos(\frac{\pi}{2} + \alpha)$ equals:

- $\sin\alpha$
- $-\sin\alpha$
- $\cos\alpha$
- $-\cos\alpha$

$\cos\alpha - \cos\beta$ equals:

- $2 \sin\frac{\alpha + \beta}{2} \sin\frac{\alpha - \beta}{2}$
- $-2 \sin\frac{\alpha + \beta}{2} \sin\frac{\alpha - \beta}{2}$
- $2 \cos\frac{\alpha + \beta}{2} \cos\frac{\alpha - \beta}{2}$
- $\cos(\alpha - \beta)$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Identity	Ayniyat	Тождество
Addition formula	Qo‘shish formulasi	Формула сложения
Double angle	Ikkilangan burchak	Двойной угол
Reduction formula	Keltirish formulasi	Формула приведения
Exact value	Aniq qiymat	Точное значение
Sum to product	Yig‘indini ko‘paytmaga	Сумма в произведение
Revision	Takrorlash	Повторение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Q1 loses marks most often for:

- arithmetic
- the sign
- the formula
- the units

Q2 must end with:

- a decimal
- the excluded values
- a graph
- a check

Q3 must end with:

- a decimal
- a rationalised surd
- a fraction
- a sketch

The best form of $\cos 2\alpha$ in Q4:

- $\cos^2\alpha - \sin^2\alpha$
- $1 - 2\sin^2\alpha$
- $2\cos^2\alpha - 1$
- any

In Q5 the first step is:

- reduce
- strip whole turns
- use parity
- use a calculator

Work on the mistakes means:

- fix the wrong line
- rewrite the solution
- copy the answer
- skip it

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\cos\alpha = -\dfrac{3}{5}$ (II): $\sin\alpha$

   ANSWER $\dfrac{4}{5}$

2. Same: $\tan\alpha$

   ANSWER $-\dfrac{4}{3}$

3. $\cos 15°$

   ANSWER $\dfrac{\sqrt{6}+\sqrt{2}}{4}$

4. $\sin 225°$

   ANSWER $-\dfrac{\sqrt{2}}{2}$

5. $\cos 750°$

   ANSWER $\dfrac{\sqrt{3}}{2}$

6. Factorise $\sin 5x - \sin x$

   ANSWER $2\cos 3x \sin 2x$

7. $\tan\alpha + \cot\alpha$

   ANSWER $\dfrac{1}{\sin\alpha\cos\alpha}$

Medium · the standard question

7 PROBLEMS

1. $\sin\alpha = \frac{5}{13}$ (I): $\sin 2\alpha$

ANSWER $\frac{120}{169}$

2. Same: $\cos 2\alpha$

ANSWER $\frac{119}{169}$

3. $\sin 15°$

ANSWER $\frac{\sqrt{6}-\sqrt{2}}{4}$

4. $\tan 15°$

ANSWER $2-\sqrt{3}$

5. Simplify $\sin(\pi-\alpha)+\cos(\frac{\pi}{2}+\alpha)$

ANSWER 0

6. Simplify $\cos^4\alpha - \sin^4\alpha$

ANSWER $\cos 2\alpha$

7. $\sin 75° + \sin 15°$

ANSWER $\frac{\sqrt{6}}{2}$

Hard · stretch and reason

7 PROBLEMS

1. Prove $\dfrac{\sin 2\alpha}{1 + \cos 2\alpha} = \tan \alpha$

   ANSWER Use $2\cos^2\alpha$

2. Prove $\sin 3\alpha = 3 \sin \alpha - 4 \sin^3\alpha$

   ANSWER $3\alpha = 2\alpha + \alpha$

3. Solve $\sin 3x + \sin x = 0$

   ANSWER $x = \dfrac{\pi k}{2}$

4. $\tan \alpha = 2$: $\tan 2\alpha$

   ANSWER $-\dfrac{4}{3}$

5. Simplify $(\sin \alpha + \cos \alpha)^2 + (\sin \alpha - \cos \alpha)^2$

   ANSWER 2

6. $\sin \alpha + \cos \alpha = \dfrac{7}{5}$: $\sin 2\alpha$

   ANSWER $\dfrac{24}{25}$

7. Prove $\cot \alpha - \tan \alpha = 2 \cot 2\alpha$

   ANSWER Common denominator, then $\sin 2\alpha$

07 Homework

Homework — 5 tasks

Rewrite in full every question that lost a mark before the next chapter begins.

1. Given $\sin \alpha = -\dfrac{12}{13}$ in quadrant III, find $\cos \alpha$ and $\tan 2\alpha$.
2. Find $\sin 105°$ exactly.

3. Evaluate $\cos 240°$ and $\sin 930°$.

4. Prove $\dfrac{1 - \cos 2\alpha}{\sin 2\alpha} = \tan \alpha$.

5. Factorise $\cos 7x + \cos 3x$ and solve $\cos 7x + \cos 3x = 0$.

Numerical sequences

A function whose domain is the counting numbers — and the two ways of describing one.

LESSON 69–71 | 3 × 40 MIN | ALGEBRA 9, §29 | IGX 11.1–11.2

LESSON CLOCK

15' | 30' | 35' | 35' | 5

What a sequence is	15 min
Two descriptions	30 min
Finding terms	35 min
Behaviour	35 min
Homework	5 min

LEARNING OBJECTIVES

- Define a sequence and use the notation a_n for its nth term.
- Describe a sequence by a formula for the nth term and by a recurrence.
- Find any term from either description, and continue a given sequence.
- Recognise increasing, decreasing, bounded and periodic sequences.

TEXTBOOK

National	pp. 167–173
Cambridge	Core & Extended, pp. 220–228
Workbook	Exercise 11.1–11.2

01 Explanation

What a sequence is

A **numerical sequence** is a list of numbers in a fixed order:

$$a_1, a_2, a_3, \dots, a_n, \dots$$

The subscript is the position, so a_1 is the first term, a_5 the fifth and a_n the nth. Order matters: $1, 2, 3$ and $3, 2, 1$ are different sequences.

SEQUENCE	FIRST FOUR TERMS	DESCRIPTION
natural numbers	$1, 2, 3, 4$	$a_n = n$
odd numbers	$1, 3, 5, 7$	$a_n = 2n - 1$
squares	$1, 4, 9, 16$	$a_n = n^2$
powers of two	$2, 4, 8, 16$	$a_n = 2^n$
alternating	$-1, 1, -1, 1$	$a_n = (-1)^n$

A SEQUENCE IS A FUNCTION

a_n is nothing but $f(n)$ with the domain restricted to $1, 2, 3, \ldots$. Everything learnt about functions in Chapter I — increasing, decreasing, bounded — applies unchanged, which is why this chapter follows that one.

Two descriptions

	FORMULA FOR THE NTH TERM	RECURRENCE
looks like	$a_n = 2n - 1$	$a_1 = 1, a_{n+1} = a_n + 2$
gives a_{100}	at once	only after 99 steps
shows the pattern	less clearly	very clearly
needs	nothing	a starting value

A **recurrence relation** says how to get from one term to the next; it is useless without the first term. $a_{n+1} = a_n + 2$ describes the odd numbers if $a_1 = 1$ and the even numbers if $a_1 = 2$.

The most famous recurrence needs two starting values:

$$F_1 = 1, F_2 = 1, F_{n+2} = F_{n+1} + F_n — 1, 1, 2, 3, 5, 8, 13, \ldots$$

CONTINUING A PATTERN IS A GUESS, NOT A DEFINITION

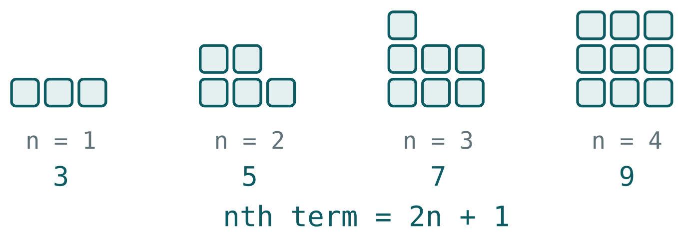
$1, 2, 4, \ldots$ may be 2 to a power, or the number of pieces a circle is cut into by n chords — which continues $8, 16, 31$. Cambridge questions say “the nth term is…” precisely to avoid this. When a question asks you to continue, give the rule you used.

Finding terms

Three standard questions, and how each is done.

QUESTION	METHOD	EXAMPLE
find a_7	substitute $n = 7$	$a_n = n^2 - 1 \Rightarrow a_7 = 48$
is 50 a term?	solve $a_n = 50$ for n	$n^2 - 1 = 50 \Rightarrow n^2 = 51$ — *not a whole number, so no*
which term is 99?	solve and keep $n \in \mathbb{N}$	$n^2 - 1 = 99 \Rightarrow n = 10$

The middle row is the one that carries marks: a number belongs to the sequence only if the equation gives a **positive whole** value of n.



The terms of a sequence plotted against their position — a set of isolated points, never a curve.

> N COUNTS, SO IT IS A POSITIVE INTEGER
>
> $n = 7.2$ or $n = -3$ means the number is not a term. Solving the equation is only half the work; checking that n is admissible is the other half.

Behaviour

NAME	MEANS	EXAMPLE
increasing	$a_{n+1} > a_n$	$1, 3, 5, 7, \ldots$
decreasing	$a_{n+1} < a_n$	$1, \frac{1}{2}, \frac{1}{3}, \ldots$
bounded above	$a_n \le M$	$1 - \frac{1}{n} \le 1$
bounded below	$a_n \ge m$	$n^2 \ge 1$
periodic	repeats after a fixed number of steps	$1, -1, 1, -1, \ldots$
constant	$a_n = c$	$5, 5, 5, \ldots$

To decide between increasing and decreasing, form $a_{n+1} - a_n$ and look at its sign — which is exactly the test used for functions in Chapter I.

BOUNDED AND MONOTONIC TOGETHER MEAN SOMETHING

A sequence that increases but never passes a ceiling must settle down towards a value. That idea — the limit — is Grade 11 work, but it is already visible in $1 - \frac{1}{n}$, which climbs forever and never reaches 1.

02 Worked examples

EXAMPLE 1 **Given $a_n = 3n - 2$, find a_1, a_5 and a_{20}.**

1. $a_1 = 3 \cdot 1 - 2 = 1$
2. $a_5 = 15 - 2 = 13$
3. $a_{20} = 60 - 2 = 58$
4. The sequence is $1, 4, 7, 10, \ldots$.

   Each term is 3 more.

ANSWER

$1, 13, 58$

EXAMPLE 2 **Is 100 a term of $a_n = n^2 - 1$?**

1. Solve $n^2 - 1 = 100$.
2. $n^2 = 101$
3. $n = \sqrt{101} \approx 10.05$ — not a whole number.
4. No: 100 is not a term.

ANSWER

No

EXAMPLE 3 **A sequence has $a_1 = 2$ and $a_{n+1} = 2a_n + 1$. Write the first five terms.**

1. $a_2 = 2 \cdot 2 + 1 = 5$
2. $a_3 = 2 \cdot 5 + 1 = 11$
3. $a_4 = 2 \cdot 11 + 1 = 23$
4. $a_5 = 2 \cdot 23 + 1 = 47$

ANSWER

$2, 5, 11, 23, 47$

03 Interactive model

Write 1, 2, 4 on the board and ask for the next term; collect the different answers, then show that all of them can be justified — which is why a rule, not a pattern, defines a sequence.

Terms from a formula

$$a_n = n^2 - 3n + 4$$

n **1** value **2**

The domain is the counting numbers, so only whole n has a term. The first four are $2, 2, 4, 8$.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Sequence	Ketma-ketlik	Последовательность
Term	Had	Член
nth term	n-had	n-й член
General term	Umumiy had	Общий член
Recurrence relation	Rekurrent munosabat	Рекуррентное соотношение
Increasing	O‘suvchi	Возрастающая
Decreasing	Kamayuvchi	Убывающая
Bounded	Chegaralangan	Ограниченная

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

a_n denotes:

- the sum
- the nth term
- the number of terms
- the first term

A recurrence relation needs:

- nothing extra
- a starting value
- a graph
- a limit

$a_n = 2n - 1$ gives $a_4 =$

- 5
- 7
- 8
- 9

n must be:

- any real number
- a positive integer
- positive
- an integer

To test increasing, look at:

- a_n
- $a_{n+1} - a_n$
- a_1
- the graph only

$1, -1, 1, -1, \ldots$ is:

- increasing
- decreasing
- periodic
- unbounded

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up   7 PROBLEMS

1. $a_n = 3n - 2$: a_1

ANSWER 1

2. Same: a_5

ANSWER 13

3. $a_n = n^2$: a_6

ANSWER 36

4. $a_n = 2^n$: a_5

ANSWER 32

5. $a_n = (-1)^n$: a_3

ANSWER -1

6. Next term of $2, 5, 8, 11, \dots$

ANSWER 14

7. Next term of $1, 4, 9, 16, \dots$

ANSWER 25

Medium · the standard question

7 PROBLEMS

1. $a_n = n^2 - 1$: a_7

ANSWER 48

2. Is 100 a term of $a_n = n^2 - 1$?

ANSWER No

3. Which term of $a_n = n^2 - 1$ is 99?

ANSWER The 10th

4. $a_1 = 2,\ a_{n+1} = 2a_n + 1$: first five terms

ANSWER $2, 5, 11, 23, 47$

5. $a_1 = 1,\ a_2 = 1,\ a_{n+2} = a_{n+1} + a_n$: a_7

ANSWER 13

6. Formula for $3, 5, 7, 9, \ldots$

ANSWER $a_n = 2n + 1$

7. Formula for $1, \frac{1}{2}, \frac{1}{3}, \ldots$

ANSWER $a_n = \frac{1}{n}$

Hard · stretch and reason

7 PROBLEMS

1. Is $a_n = \dfrac{n}{n+1}$ increasing or decreasing?

   ANSWER Increasing

2. Is it bounded?

   ANSWER Yes — between $\dfrac{1}{2}$ and 1

3. Formula for $2, 6, 12, 20, 30, \ldots$

   ANSWER $a_n = n(n+1)$

4. Formula for $1, -2, 3, -4, \ldots$

   ANSWER $a_n = (-1)^{n+1} n$

5. Which term of $a_n = n^2 + n$ equals 132?

   ANSWER The 11th

6. Show $a_n = 1 - \dfrac{1}{n}$ never reaches 1

   ANSWER $\dfrac{1}{n} > 0$ for all n

7. First term of $a_n = n^2 - 20n$ that is positive

   ANSWER $a_{21} = 21$

07 Homework

Homework — 5 tasks

When you continue a sequence, always state the rule you used.

1. Given $a_n = 4n - 3$, write the first five terms and find a_{50}.
2. Is 81 a term of $a_n = 2n + 1$? Which one?
3. A sequence has $a_1 = 3$ and $a_{n+1} = 3a_n - 2$. Write four terms.

4. Find a formula for the nth term of $5, 8, 11, 14, \ldots$.

5. Say whether $a_n = \dfrac{n + 1}{n}$ is increasing or decreasing, and whether it is bounded.

The arithmetic progression

Add the same number every time, and the nth term becomes a straight line in n.

LESSON 72–74 | 3 × 40 MIN | ALGEBRA 9, §30 | IGX 11.3

LESSON CLOCK

18' | 32' | 35' | 30' | 5

The definition	18 min
The nth term	32 min
The middle-term property	35 min
Applications	30 min
Homework	5 min

LEARNING OBJECTIVES

- Define an arithmetic progression and find its common difference.
- Use $a_n = a_1 + (n - 1)d$ in all four directions.
- Use the characteristic property $2a_n = a_{n-1} + a_{n+1}$.
- Insert arithmetic means between two given numbers.

TEXTBOOK

National	pp. 174–180
Cambridge	Core & Extended, pp. 229–234
Workbook	Exercise 11.3

01 Explanation

The definition

A sequence is an **arithmetic progression** if each term after the first is obtained by adding the same number d, called the **common difference**.

$$a_{n+1} = a_n + d \text{ equivalently } d = a_{n+1} - a_n$$

SEQUENCE	A_1	D	BEHAVIOUR
$3, 7, 11, 15, \ldots$	3	4	increasing
$20, 17, 14, 11, \ldots$	20	-3	decreasing
$5, 5, 5, 5, \ldots$	5	0	constant
$1, 2, 4, 8, \ldots$	1	—	not an AP

CHECK THE DIFFERENCE AT LEAST TWICE

> $1, 3, 5, 9$ passes the test between the first pair and fails at the last. To prove a sequence is arithmetic you must show $a_{n+1} - a_n$ is the same for **every** n — in practice, by computing the difference in general, not on two examples.

The nth term

Starting at a_1 and adding d a total of $n - 1$ times:

$$a_n = a_1 + (n - 1)d$$

GIVEN	FIND	HOW
a_1, d, n	a_n	substitute
a_1, d, a_n	n	solve a linear equation
a_1, a_n, n	d	$d = \dfrac{a_n - a_1}{n - 1}$
two terms	a_1, d	two equations, two unknowns

The last row is the standard examination question. From $a_3 = 11$ and $a_8 = 31$: subtracting gives $5d = 20$, so $d = 4$ and $a_1 = 11 - 2 \cdot 4 = 3$.

> IT IS A STRAIGHT LINE
>
> $a_n = dn + (a_1 - d)$ is linear in n, with gradient d. Plotting the terms gives points on a straight line — which is why the difference of two terms is always a multiple of d, and why $n - 1$, not n, appears in the formula.

The middle-term property

Every term except the first is the average of its two neighbours:

$$a_n = \frac{a_{n-1} + a_{n+1}}{2} \quad \text{that is } 2a_n = a_{n-1} + a_{n+1}$$

This is the **characteristic property**: a sequence is arithmetic **if and only if** it holds for every $n > 1$. It is therefore the standard way to prove that something is an AP, and the standard way to fill a gap.

QUESTION	USING THE PROPERTY
$7, x, 19$	$2x = 26 \Rightarrow x = 13$
$x, 12, 3x$	$24 = x + 3x \Rightarrow x = 6$
is $\log 2, \log 4, \log 8$ an AP?	$2 \log 4 = \log 16 = \log 2 + \log 8$

WHY IT IS CALLED THE ARITHMETIC MEAN

$\frac{a+b}{2}$ is the arithmetic mean of a and b — the same quantity that appeared in the inequality $\frac{a+b}{2} \geq \sqrt{ab}$ last quarter. The progression is named after the mean, not the other way round.

Applications

Inserting means. To put k terms between a and b so that the whole list is an AP, there are $k + 1$ steps from a to b, so $d = \frac{b-a}{k+1}$.

Counting terms. How many multiples of 7 lie between 100 and 500? The first is 105, the last 497, and $497 = 105 + (n - 1) \cdot 7$ gives $n = 57$.

SITUATION	THE AP
a salary rising by a fixed amount	$a_1 = \text{start}, d = \text{rise}$
seats increasing by 2 per row	$d = 2$
a stack of logs, one fewer each layer	$d = -1$
simple interest	d is the annual interest

N MUST COME OUT A POSITIVE WHOLE NUMBER

In a counting question a fractional n means the number you chose is not a term of the progression — usually because the first or last term was picked wrongly. Check both ends before dividing.

02 Worked examples

EXAMPLE 1 **In an AP, $a_3 = 11$ and $a_8 = 31$. Find a_1, d and a_{20}.**

1. $a_8 - a_3 = 5d = 20$

   Five steps apart.

2. $d = 4$

3. $a_1 = a_3 - 2d = 11 - 8 = 3$

4. $a_{20} = 3 + 19 \cdot 4 = 79$

ANSWER

$a_1 = 3, d = 4, a_{20} = 79$

EXAMPLE 2 **How many multiples of 7 lie between 100 and 500?**

1. First: 105. Last: 497.

2. $497 = 105 + (n - 1) \cdot 7$

3. $392 = 7(n - 1) \Rightarrow n - 1 = 56$

4. $n = 57$

ANSWER

57

EXAMPLE 3 **Insert four numbers between *2* and *17* so that all six form an AP.**

1. There are 5 steps from 2 to 17.
   $k + 1 = 5.$

2. $d = \dfrac{15}{5} = 3$

3. The terms: $2, 5, 8, 11, 14, 17$.

4. Check: the difference is 3 throughout ✓

ANSWER

$5, 8, 11, 14$

03 Interactive model

Ask why the formula has n − 1 rather than n; getting the class to count the steps from a_1 to a_4 on their fingers prevents the commonest error of the whole chapter.

a_1, d and the nth term

$$a_n = 5 + 3(n - 1)$$

n **1** | value **5**

A straight line in n: the first term is 5 and every step adds 3.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Arithmetic progression	Arifmetik progressiya	Арифметическая прогрессия
Common difference	Ayirma	Разность
First term	Birinchi had	Первый член
Characteristic property	Xarakteristik xossa	Характеристическое свойство
Arithmetic mean	O‘rta arifmetik	Среднее арифметическое
Insert	Qo‘yish	Вставить
Finite	Chekli	Конечная
Infinite	Cheksiz	Бесконечная

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

In an AP, d equals:

- $a_n a_{n+1}$
- $a_{n+1} - a_n$
- $\frac{a_{n+1}}{a_n}$
- a_1

a_n equals:

- $a_1 + nd$
- $a_1 + (n-1)d$
- $a_1 d^{n-1}$
- $a_1 - nd$

$3, 7, 11, \ldots$ has $a_{10} =$

- 39
- 40
- 43
- 47

The characteristic property is:

- $a_n^2 = a_{n-1}\, a_{n+1}$
- $2a_n = a_{n-1} + a_{n+1}$
- $a_n = a_1 + d$
- $a_n = nd$

To insert 4 terms between 2 and 17, d is:

- 3
- $\frac{15}{4}$
- 4
- 5

$20, 17, 14, \ldots$ is:

- increasing
- decreasing
- constant
- not an AP

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. d of $3, 7, 11, \ldots$

ANSWER 4

2. d of $20, 17, 14, \ldots$

ANSWER -3

3. $a_1 = 5, d = 3$: a_4

ANSWER 14

4. $a_1 = 2, d = 6$: a_{10}

ANSWER 56

5. $3, 7, 11, \ldots$: a_{10}

ANSWER 39

6. Middle term of $7, x, 19$

ANSWER 13

7. Is $5, 5, 5, \ldots$ an AP?

ANSWER Yes, $d = 0$

Medium · the standard question

7 PROBLEMS

1. $a_3 = 11, a_8 = 31$: d

ANSWER 4

2. Same: a_1

ANSWER 3

3. Same: a_{20}

ANSWER 79

4. $a_1 = 4, a_n = 49, d = 5$: n

ANSWER 10

5. Insert four numbers between 2 and 17

ANSWER $5, 8, 11, 14$

6. Multiples of 7 between 100 and 500

ANSWER 57

7. Find x if $x, 12, 3x$ is an AP

ANSWER 6

Hard · stretch and reason

7 PROBLEMS

1. $a_5 = 17, a_{12} = 45$: a_1

ANSWER 1

2. Which term of $7, 11, 15, \ldots$ equals 107?

ANSWER The 26th

3. Is 200 a term of $3, 7, 11, \ldots$?

ANSWER No — n is not whole

4. Three numbers in AP have sum 21 and product 231

ANSWER $3, 7, 11$

5. How many two-digit multiples of 4 are there?

ANSWER 22

6. In an AP, $a_4 + a_9 = 30$. Find $a_6 + a_7$

ANSWER 30

7. Prove $\log 2, \log 4, \log 8$ is an AP

ANSWER $2 \log 4 = \log 2 + \log 8$

07 Homework

Homework — 5 tasks

Write a_1 and d at the top of every solution before doing anything else.

1. In an AP, $a_1 = 6$ and $d = 5$. Find a_{15} and say which term equals 106.
2. Given $a_4 = 14$ and $a_9 = 34$, find a_1 and d.
3. Insert three numbers between 4 and 24 to make an AP.
4. How many multiples of 6 lie between 50 and 400?
5. Find x if $2x - 1, x + 4, 3x + 1$ form an AP.

The sum of the first n terms of an arithmetic progression

Gauss pairs the ends, and a sum of a hundred terms becomes one multiplication.

LESSON 75–76 | 2 × 40 MIN | ALGEBRA 9, §31 | IGX 11.4

LESSON CLOCK

16' | 20' | 24' | 16' | 4'

Gauss's trick	16 min
Two forms	20 min
Solving for n	24 min
Word problems	16 min
Homework	4 min

LEARNING OBJECTIVES

- Derive $S_n = n(a_1 + a_n)/2$ by writing the sum forwards and backwards.
- Use both forms of the formula and choose the right one.
- Solve for n when the sum is given, rejecting inadmissible roots.
- Apply the formula to counting and word problems.

TEXTBOOK

National	pp. 181–187
Cambridge	Core & Extended, pp. 235–240
Workbook	Exercise 11.4

01 Explanation

Gauss's trick

Write the sum forwards, and underneath it the same sum backwards:

$$S_n = a_1 + a_2 + \ldots + a_{n-1} + a_n$$

$$S_n = a_n + a_{n-1} + \ldots + a_2 + a_1$$

Add the two lines column by column. In an AP each column has the same total, because going one step forward in the top row is matched by one step back in the bottom one:

$$2S_n = n(a_1 + a_n) \Longrightarrow S_n = \frac{n(a_1 + a_n)}{2}$$

THE SCHOOLBOY AND THE HUNDRED NUMBERS

The story that Gauss summed 1 to 100 in seconds by pairing $1 + 100$, $2 + 99$, … is exactly this proof: fifty pairs, each worth 101, giving 5050. The formula above is that idea written for any progression.

Two forms

Substituting $a_n = a_1 + (n - 1)d$ into the first form gives the second:

$$S_n = \frac{n(a_1 + a_n)}{2} \qquad S_n = \frac{n[2a_1 + (n-1)d]}{2}$$

USE THE FIRST WHEN	USE THE SECOND WHEN
the last term is given	the common difference is given
the ends are known	only the start and the step are known
counting a finished list	n is the unknown

Example. $a_1 = 3$, $d = 4$, $n = 20$. The second form gives $S_{20} = \frac{20(6 + 19 \cdot 4)}{2} = 10 \cdot 82 = 820$.

DO NOT MIX THE TWO FORMS

$\frac{n(2a_1 + a_n)}{2}$ is neither of them and is always wrong. Decide which form you are using, write it out in full, and only then substitute.

Solving for n

The second form is a **quadratic in n**, so a question that gives S_n and asks for n produces a quadratic equation — and Chapter I comes back into use.

Example. $a_1 = 3$, $d = 4$, $S_n = 210$. Then

$$\frac{n[6 + 4(n-1)]}{2} = 210 \Longrightarrow n(4n + 2) = 420 \Longrightarrow 2n^2 + n - 210 = 0$$

The roots are $n = 10$ and $n = -10.5$. A number of terms cannot be negative or fractional, so $n = 10$.

REJECTING THE SECOND ROOT IS PART OF THE ANSWER

Writing "$n = 10$ or $n = -10.5$" loses a mark. Write $n = 10$, and one short sentence saying that the other root is rejected because n counts terms.

Word problems

PROBLEM	THE PROGRESSION	ANSWER
$1 + 2 + \ldots + 100$	$a_1 = 1, a_n = 100, n = 100$	5050
sum of the first n odd numbers	$a_1 = 1, d = 2$	n^2
a theatre: 20 seats in row 1, 2 more each row, 15 rows	$a_1 = 20, d = 2, n = 15$	510
saving 100 the first month, 50 more each month, for a year	$a_1 = 100, d = 50, n = 12$	4500

The second row is worth remembering on its own: the sum of the first n odd numbers is exactly n^2. It is the neatest small theorem in the chapter, and it can be seen at once by arranging counters in a square.

NAME A_1, D AND N FIRST

Almost every word problem in this section is solved the moment those three numbers are written down. Extracting them from the sentence is the whole difficulty; the arithmetic afterwards is one line.

02 Worked examples

EXAMPLE 1 **Find S_{20} for the AP $3, 7, 11, \ldots$.**

1. $a_1 = 3, d = 4, n = 20$
2. $S_{20} = \dfrac{20[6 + 19 \cdot 4]}{2}$

   The second form.
3. $= 10 \cdot 82$
4. $= 820$

ANSWER

820

EXAMPLE 2 **For $a_1 = 3, d = 4$, how many terms give a sum of 210?**

1. $\dfrac{n[6 + 4(n - 1)]}{2} = 210$
2. $n(4n + 2) = 420 \Rightarrow 2n^2 + n - 210 = 0$
3. $n = 10$ or $n = -10.5$.
4. $n = 10$; the other root is rejected.

   n counts terms.

ANSWER

$n = 10$

EXAMPLE 3 **A theatre has *20* seats in the first row and *2* more in each next row. How many seats in *15* rows?**

1. $a_1 = 20, d = 2, n = 15$

2. $a_{15} = 20 + 14 \cdot 2 = 48$

3. $S_{15} = \dfrac{15(20 + 48)}{2}$

   The first form.

4. $= \dfrac{15 \cdot 68}{2} = 510$

ANSWER

510 seats

03 Interactive model

Ask the class to add 1 to 100 before showing any formula; the pairing they invent is the proof, and they will not forget a formula they built.

S_n as n grows

$$S_n = \frac{n(7 + 3n)}{2}$$

n **1**

value **5**

The sum of $5, 8, 11, \ldots$. It grows like n^2, not like n — doubling n roughly quadruples the sum.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Sum of n terms	n ta hadning yig‘indisi	Сумма n членов
Partial sum	Qismiy yig‘indi	Частичная сумма
Pairing	Juftlash	Разбиение на пары
Quadratic in n	n ga nisbatan kvadrat	Квадратное относительно n
Admissible root	Joiz ildiz	Допустимый корень
Reject	Rad etish	Отбросить
Word problem	Matnli masala	Текстовая задача
Total	Jami	Итого

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

S_n equals:

- $\dfrac{n(a_1 + a_n)}{2}$
- $n(a_1 + a_n)$
- $\dfrac{a_1 + a_n}{2}$
- na_n

The other form is:

- $\dfrac{n[2a_1 + nd]}{2}$
- $\dfrac{n[2a_1 + (n - 1)d]}{2}$
- $\dfrac{n[a_1 + d]}{2}$
- $na_1 + d$

$1 + 2 + \ldots + 100$ equals:

- 5000
- 5050
- 5100
- 10100

The sum of the first n odd numbers is:

- n
- n^2
- $2n$
- $n(n+1)$

Given S_n, finding n needs:

- a linear equation
- a quadratic equation
- a graph
- a calculator

A negative root for n is:

- kept
- rejected
- halved
- squared

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. S_{10} of $1, 2, 3, \dots$

   ANSWER 55

2. $1 + 2 + \dots + 100$

   ANSWER 5050

3. S_5 of $2, 5, 8, \dots$

   ANSWER 40

4. $a_1 = 3, d = 4, n = 20$: S_{20}

   ANSWER 820

5. Sum of the first 7 odd numbers

   ANSWER 49

6. S_4 of $10, 8, 6, \dots$

   ANSWER 28

7. $a_1 = 1, a_n = 19, n = 10$: S_n

   ANSWER 100

Medium · the standard question

7 PROBLEMS

1. $a_1 = 5, d = 3, n = 12$: S_{12}

ANSWER *258*

2. $a_1 = 3, d = 4, S_n = 210$: n

ANSWER *10*

3. Sum of all two-digit numbers

ANSWER *4905*

4. Sum of multiples of 7 below 100

ANSWER *735*

5. A theatre: 20 seats, $+2$ a row, 15 rows

ANSWER *510*

6. Saving 100, then $+50$ a month, for 12 months

ANSWER *4500*

7. S_{20} of $50, 47, 44, \ldots$

ANSWER *430*

Hard · stretch and reason

7 PROBLEMS

1. $a_3 = 11, a_8 = 31$: S_{15}

   ANSWER 465

2. How many terms of $2, 5, 8, \dots$ give a sum of 155?

   ANSWER 10

3. $S_5 = 40$ and $S_{10} = 155$: find a_1 and d

   ANSWER $a_1 = 2, d = 3$

4. Sum of the odd numbers between 1 and 99

   ANSWER 2500

5. Prove the sum of the first n odd numbers is n^2

   ANSWER $\frac{n(1 + 2n - 1)}{2}$

6. A stack: 30 logs at the bottom, one fewer each layer, 20 layers

   ANSWER 410

7. For which n is S_n of $20, 17, 14, \dots$ greatest?

   ANSWER $n = 7, S_7 = 77$

07 Homework

Homework — 5 tasks

State which of the two forms you are using before substituting anything.

1. Find S_{15} for the AP $4, 9, 14, \dots$.
2. Find the sum of all multiples of 5 between 1 and 200.
3. For $a_1 = 2$ and $d = 3$, how many terms give a sum of 155?
4. A row of trees: 12 in the first row, 3 more in each next, 10 rows. How many trees?

5. Given $S_4 = 26$ and $S_8 = 100$, find a_1 and d.

Control work 6, and the quarter reviewed

Trigonometric formulae and arithmetic progressions in one paper, before the third quarter closes.

LESSON 77–78 | 2 × 40 MIN | ALGEBRA 9, NAZORAT ISHI 6 | IGX 11 REVIEW

LESSON CLOCK

3 | 40' | 12' | 20' | 5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The quarter	5 min

LEARNING OBJECTIVES

- Find a term and a sum of an arithmetic progression under time.
- Solve for n from a given sum, rejecting the inadmissible root.
- Apply a trigonometric formula from Chapter III accurately.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 167–187
Cambridge	Core & Extended, pp. 220–240
Workbook	Control paper A6

01 Explanation

The paper — 30 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Write the first four terms of $a_n = 3n + 1$ and of $a_1 = 2, a_{n+1} = 2a_n - 1$	4	L69–71
2	In an AP, $a_4 = 14$ and $a_9 = 34$: find a_1 and d	5	L72–74
3	Find S_{15} for $4, 9, 14, \dots$	5	L75–76
4	For $a_1 = 2, d = 3$, find n with $S_n = 155$	6	L75–76
5	Evaluate $\sin 150° + \cos 300°$	5	L61–62
6	$\sin\alpha = \dfrac{3}{5}$ (I): find $\sin 2\alpha$	5	L59–60

WHERE THE MARKS ACTUALLY GO

Q2 carries one mark for using the gap of five steps rather than solving two full equations; Q4 two marks for forming the quadratic and one for rejecting the negative root; Q5 one for stating the reduction used; Q6 one for finding $\cos\alpha$ before substituting.

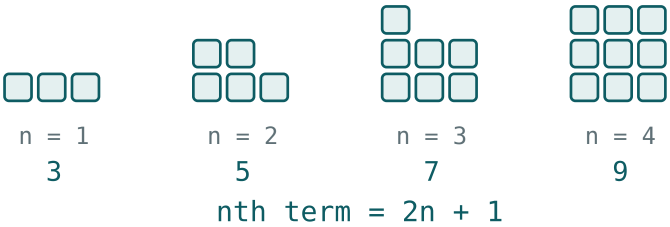
Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
recurrence applied to n instead of a_n	$2, 3, 5, 7$	$a_4 = 2 \cdot 5 - 1 = 9$
n	$a_1 + nd$	$a_1 + (n - 1)d$
forms of S_n mixed	$\frac{n(2a_1 + a_n)}{2}$	pick one form and write it out
negative root kept	$n = 10 \text{ or } -10.3$	reject, and say why
reduction sign wrong	$\cos 300° = -\frac{1}{2}$	quadrant IV: cosine is positive
$\sin 2\alpha = 2\sin\alpha$	$\frac{6}{5}$	$2\sin\alpha\cos\alpha$
no units or no sentence	510	“510 seats”

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The quarter in one page

BLOCK	THE SENTENCE
relations	one function and a quadrant give all four
identities	one side only, and head for sines and cosines
parity and period	strip whole turns, then use $-\alpha$
addition formulae	sine mixes and keeps the sign; cosine matches and flips it
double angle	set $\beta = \alpha$; three faces of $\cos 2\alpha$
reduction	two questions: does the name change, and what is the sign?
sum to product	half-sum and half-difference, so that you can factorise
sequences	a function of n, given by a formula or a recurrence
arithmetic progression	$a_n = a_1 + (n - 1)d$
its sum	$S_n = \dfrac{n(a_1 + a_n)}{2}$ — *pair the ends*



An arithmetic progression plotted: the terms lie on a straight line of gradient d.

LOOKING FORWARD

Quarter IV keeps the same shape and changes the operation: the geometric progression multiplies where the arithmetic one adds. Everything — the nth term, the characteristic property, the sum — has a geometric twin, and the last part of the year adds probability.

02 Worked examples

EXAMPLE 1 **Model answer, Q2: $a_4 = 14, a_9 = 34$.**

1. $a_9 - a_4 = 5d = 20$

   Five steps apart.

2. $d = 4$

3. $a_1 = a_4 - 3d = 14 - 12 = 2$

4. $a_1 = 2, d = 4$

ANSWER

$a_1 = 2, d = 4$

EXAMPLE 2 **Model answer, Q4: $a_1 = 2, d = 3, S_n = 155$.**

1. $\dfrac{n[4 + 3(n - 1)]}{2} = 155$

2. $n(3n + 1) = 310 \Rightarrow 3n^2 + n - 310 = 0$

3. $n = 10$ or $n = -\dfrac{31}{3}$.

4. $n = 10$; the negative root is rejected.

   n counts terms.

ANSWER

$n = 10$

EXAMPLE 3 **Model answer, Q5: evaluate $\sin 150° + \cos 300°$.**

1. $\sin 150° = \sin(180° - 30°) = \sin 30°$

   Name stays; quadrant II.

2. $= \frac{1}{2}$

3. $\cos 300° = \cos(360° - 60°) = \cos 60°$

   Quadrant IV: cosine positive.

4. $\frac{1}{2} + \frac{1}{2} = 1$

ANSWER

1

03 Interactive model

Mark Q4 aloud: read out a solution that keeps both roots, and let the class say what the sentence about n counting terms is worth.

Quarter III in twelve questions

$1 + tan^2\alpha$ equals:

- $\dfrac{1}{\cos^2\alpha}$
- $\dfrac{1}{\sin^2\alpha}$
- $cos^2\alpha$
- 1

$cos(-\alpha)$ equals:

- $-cos\ \alpha$
- $cos\ \alpha$
- $sin\ \alpha$
- $-sin\ \alpha$

$sin(\alpha - \beta)$ equals:

- $sc + cs$
- $sc - cs$
- $cc - ss$
- $cc + ss$

$\cos 2\alpha$ in terms of $\sin \alpha$:

- $1 - 2\sin^2\alpha$
- $2\sin^2\alpha - 1$
- $1 + 2\sin^2\alpha$
- $2 - \sin^2\alpha$

$\sin(\pi - \alpha)$ equals:

- $\sin \alpha$
- $-\sin \alpha$
- $\cos \alpha$
- $-\cos \alpha$

$\sin \alpha + \sin \beta$ equals:

- $2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$
- $2 \cos \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$
- $\sin(\alpha + \beta)$
- $2 \sin \alpha \sin \beta$

A recurrence needs:

- nothing
- a starting value
- a limit
- a graph

a_n of an AP equals:

- $a_1 + nd$
- $a_1 + (n - 1)d$
- $a_1 d^n$
- nd

The characteristic property of an AP:

- $a_n^2 = a_{n-1} a_{n+1}$
- $2a_n = a_{n-1} + a_{n+1}$
- $a_n = nd$
- $a_n = a_1 d$

S_n equals:

- $\dfrac{n(a_1 + a_n)}{2}$
- $n(a_1 + a_n)$
- $\dfrac{a_1 + a_n}{2}$
- $n\,a_n$

$1 + 2 + \ldots + 100$:

- *5000*
- *5050*
- *5100*
- *10000*

A fractional root for n is:

- kept
- rounded
- rejected
- doubled

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Arithmetic progression	Arifmetik progressiya	Арифметическая прогрессия
Common difference	Ayirma	Разность
Sum of n terms	n ta hadning yig‘indisi	Сумма n членов
Recurrence	Rekurrent munosabat	Рекуррентное соотношение
Admissible root	Joiz ildiz	Допустимый корень
Word problem	Matnli masala	Текстовая задача
Revision	Takrorlash	Повторение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

In Q2 the fastest route is:

- two full equations
- the gap of five steps
- a graph
- trial and error

Q4 produces:

- a linear equation
- a quadratic equation
- an identity
- an inequality

The negative root in Q4 is:

- kept
- rejected with a reason
- rounded
- halved

$\cos 300°$ is:

- positive
- negative
- zero
- undefined

Q6 needs, before substituting:

- $\tan\alpha$
- $\cos\alpha$
- $\cot\alpha$
- nothing

Work on the mistakes means:

- fix the wrong line
- rewrite the solution
- copy the answer
- skip it

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $a_n = 3n + 1$: first four terms

   ANSWER $4, 7, 10, 13$

2. $a_1 = 2$, $a_{n+1} = 2a_n - 1$: first four

   ANSWER $2, 3, 5, 9$

3. $a_4 = 14$, $a_9 = 34$: d

   ANSWER 4

4. Same: a_1

   ANSWER 2

5. S_{15} of $4, 9, 14, \ldots$

   ANSWER 585

6. $\sin 150°$

   ANSWER $\frac{1}{2}$

7. $\cos 300°$

   ANSWER $\frac{1}{2}$

Medium · the standard question

7 PROBLEMS

1. $a_1 = 2, d = 3, S_n = 155$: n

ANSWER 10

2. $\sin 150° + \cos 300°$

ANSWER 1

3. $\sin \alpha = \frac{3}{5}$ (I): $\sin 2\alpha$

ANSWER $\frac{24}{25}$

4. Same: $\cos 2\alpha$

ANSWER $\frac{7}{25}$

5. Sum of all two-digit multiples of 4

ANSWER 1188

6. S_{10} of $50, 47, 44, \dots$

ANSWER 365

7. Which term of $4, 9, 14, \dots$ is 104?

ANSWER The 21st

Hard · stretch and reason

7 PROBLEMS

1. $S_4 = 26, S_8 = 100$: a_1 and d

ANSWER $a_1 = 2, d = 3$

2. Three numbers in AP: sum 15, product 80

ANSWER $2, 5, 8$

3. Evaluate $\sin 225° + \cos 315° + \tan 135°$

ANSWER -1

4. Prove $\dfrac{\sin 2\alpha}{1 + \cos 2\alpha} = \tan \alpha$

ANSWER Use $2\cos^2\alpha$

5. For which n is S_n of $30, 27, 24, \ldots$ greatest?

ANSWER $n = 10$ or 11, $S_n = 165$

6. Factorise $\cos 5x + \cos x$ and solve $= 0$

ANSWER $2 \cos 3x \cos 2x = 0$

7. Sum of the first n even numbers

ANSWER $n(n + 1)$

07 Homework

Homework — 5 tasks

Rewrite in full every question that lost a mark; the holiday is not the place to leave them.

1. In an AP, $a_5 = 20$ and $a_{12} = 48$. Find a_1, d and S_{12}.
2. For $a_1 = 3$ and $d = 5$, find n such that $S_n = 255$.
3. Evaluate $\cos 210° + \sin 330°$.
4. Given $\cos \alpha = \dfrac{4}{5}$ in quadrant IV, find $\sin 2\alpha$ and $\cos 2\alpha$.

5. Find the sum of all multiples of 3 between 1 and 100.